Joint Strategic Geometry on Decision and Objective Spaces

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Tenets and Contracts

Standing Hypotheses (H1–H6).

(H1) Decision space Γ is smooth/connected; symmetry by Lie group when needed.

(H2) G is Whitney-stratified; dynamics specified per stratum and prolonged (Filippov) at frontiers.

(H3) Classical identities hold stratumwise; frontier crossings induce a defect budget per crossing.

(H4) Evaluation functionals are Borel and L^2 under invariant/windowed laws.

(H5) EF neighborhood at independence with Fisher $F \succ 0$ and Hessian-Lipschitz L_ψ.

(H6) Controllers are Dirac-projected and band-limited PR on $[\omega_L, \omega_H]$ with margin ϵ.

Symbol Registry (standing constants and moduli)

*All constants are understood as **uniform on compact sublevel sets** of the Lyapunov functions in use.* For quick cross-references elsewhere, use `\constref{...}`. *[Symbol Registry table omitted in this build due to a tabularx parsing error; will be restored after CI fix.]*

Acceptance checks for this part.

- Frontier events: logs include N_{cross} and $t/\tau_{\min}$ bound.
- Defect budgets: per-crossing contribution tracked using $\kappa_{\max}, \theta_{\min}, r_{\mathrm{tube}}$.
- PR certificates: band-limited positive-real margin reported ($\geq \epsilon$).
- DRO dual: windowed $\rho = c\,\hat{L}\,\tau$ with $\eta^\star$ recorded.

Contents

1 Introduction and Motivation

Purpose. We develop a rigorously testable framework for coupled decision–objective dynamics. Agents evolve on a decision space Γ while a co-evolving structural environment G reshapes feasible actions and payoffs. Our objective is an *engineering-grade* interlingua that: (i) separates modeling layers by explicit contracts; (ii) states hypotheses at proof/implementation granularity; (iii) provides local theorems that are falsifiable on benchmarks; and (iv) degrades gracefully outside their stated radii.

Tensions we resolve. We make precise choices where earlier drafts blended: (a) smooth vs. stratified geometry on G; (b) algebraic identities vs. their discrete/stratified counterparts; (c) stationary vs. drifting laws; (d) geometric control vs. numerics. We control the resulting defects by No-Zeno dwell times, geometry-scaled frontier budgets, band-limited approximations, and dependence-aware estimation.

Problem statement

For a coupled process $X_t = (\gamma_t, G_t)$ on $X = \Gamma \times G$, we seek:

(P1) **Hybrid calculus.** A stratified exterior calculus on $T\Gamma \oplus TG$ that reduces to the smooth theory on each stratum and yields integrable, geometry-controlled defect terms at frontiers with No-Zeno crossings.

(P2) **Statistical semantics.** Laws (invariant or windowed) under which evaluation functionals are Borel/L^2, and a local Fisher-invariant equivalence between mixed cumulants and relative entropy with an explicit radius.

(P3) **Closure mechanisms.** A Wasserstein–KL DRO (with stratified reference λ) and a Hopf-algebraic projection that preserve symmetries and bound higher orders.

(P4) **Geometry-aware control.** A Dirac (power-preserving) projection Π and band-limited fractional controllers verifiably positive-real on their band, implying ISS under frontier defects.

(P5) **Compositional semantics.** Lipschitz functors in the FIM-normed cumulant metric with monoidal submultiplicativity, and non-expansive projections for Dirac/Hopf closures.

Modeling layers and contracts

We isolate arguments by layer. Each result carries a tag `[Rep/Alg/Stoch/Inf/Clos/Ctrl/Comp]` and ends with *Uses:* (C-codes).

Representation (Rep).
Γ smooth; G Whitney-stratified with Filippov prolongation at frontiers and strict transversality. Contract: dwell $\tau_{\min} > 0$ and frontier tubes of radius r_{tube} (Lemma 1).

Algebra (Alg).
Bigraded exterior calculus on $T\Gamma \oplus TG$; classical identities hold stratumwise; Jacobi fails by a distributional frontier term with per-crossing budget C_{fr} (Lemmas 2–3).

Stochastic (Stoch).
Occupation/invariant or windowed laws; evaluation functionals are Borel and L^2. Absolute continuity is taken *stratumwise* w.r.t. a refinement-invariant reference λ (Def. 4).

Inference (Inf).
EF neighborhood at independence with finite, whitened dictionary; Fisher $F \succ 0$. Local KL–cumulant equivalence on a ball of radius $r \le \epsilon / L_\psi$ with conditioning factor χ.

Closure (Clos).
Wasserstein–KL DRO with soft-transport dual φ_η; inner optimizer $\eta^\star$ satisfies KKT under a dominated-convergence envelope. Adaptive radius $\rho_t = c\,\widehat{L}_t \tau$ with guarantees.

Control (Ctrl).
Dirac projection Π from rank-regular constraints; “freeze on crossing, reproject after” rule. Band-limited fractional controller with KYP/phase PR certificates; passivity $\Rightarrow$ ISS with an explicit frontier term proportional to $t/\tau_{\min}$.

Composition (Comp).
Algebra-first quasi-shuffle $\odot$ (associative/commutative), *then* whitening; invariant-whitening assumption ensures symmetry equivariance. Pushforward–cumulant Lipschitz lemma justifies non-expansiveness of Dirac/Hopf projections in the FIM metric; monoidal Lipschitz constants multiply.

Standing Hypotheses (H1–H6).

(H1) Γ is a connected smooth manifold; when symmetry matters we take a Lie group (e.g. $SE(3)$).

(H2) G is a Whitney-stratified pseudomanifold $\bigsqcup_i M_i$; dynamics are specified stratumwise and prolonged by Filippov regularization. Crossings are non-explosive and strictly transversal.

(H3) On each stratum, mixed wedge, interior contraction, and Schouten–Nijenhuis bracket obey the smooth identities; frontier crossings induce a defect Δ_{fr} with $\int_0^T \|\Delta_{\mathrm{fr}}(x_t)\|dt \le C_{\mathrm{fr}} N_{\mathrm{cross}}(0,T)$.

(H4) Test functionals $T_{u,v} = \iota_{u\otimes v}\hat{S}$ are Borel/L^2 under the laws considered (invariant or windowed). Laws satisfy $P_\theta \ll \lambda$ stratumwise, with λ the stratified reference measure (Def. 4).

(H5) In an EF neighborhood of independence $P_0 = P_\Gamma \otimes P_G$, a finite statistic dictionary has $F = \mathrm{FIM}(0) \succ 0$ and $\nabla^2\psi$ is L_ψ-Lipschitz on a ball around 0.

(H6) Controllers use Dirac projections and band-limited positive-real fractional operators; entropy-aware gain adaptation stays within a PR-admissible set or evolves on a slower time scale than the plant.

Non-goals and scope

We do not claim global equivalences far from independence; we do not develop an unconstrained discrete multivector calculus on arbitrary meshes; we restrict to band-limited fractional realizations; and we avoid unbenchmarked claims.

Repository style and isolation

Every theorem/lemma is tagged by layer and ends with a *Uses:* line that lists the C-codes it touches from the symbol registry. Cross-references name the exact contracts invoked.

Notation

Bidegrees use (p, q); total grade is $p + q$. Pairings $\langle\cdot,\cdot\rangle$ are induced by the relevant metric; FIM norms are $\|z\|_{\mathrm{FIM}^{-1}} = \|z\|_{\mathrm{FIM}^{-1}}$.

Roadmap

Section 2 fixes the hybrid calculus, proving No-Zeno (Lemma 1) and a geometry-scaled defect (Lemma 3). Section 3 sets stochastic semantics (existence, windowing, measurability). Section 4 introduces the stratified reference λ and the soft-transport dual with KKT and an adaptive ρ_t. Section 5 states the Fisher-invariant KL–cumulant theorem with explicit radius and conditioning corollary. Section 6 builds Dirac projections and PR-certified fractional control, then proves passivity $\Rightarrow$ ISS with a frontier term scaling as $t/\tau_{\min}$. Section 7 develops SVD plane selection with J-structure and DEC-compatible shape derivatives with a frontier freeze. Section 8 adopts the *algebra-first* quasi-shuffle, an *invariant* whitening assumption, and a pushforward–cumulant Lipschitz lemma yielding non-expansiveness in the FIM metric with monoidal submultiplicativity. Section 9 standardizes dependence-aware CIs, model-order selection, and FDR control. Section 10 closes with executable benchmarks, negative controls, and degeneracy stress tests.

2 State Spaces and Hybrid Exterior Algebra

This section fixes the geometric setting and proves the *Hybrid Calculus Axioms*. New here: (i) a *No-Zeno* lemma under strict transversality with an explicit dwell time $\tau_{\min}$ (Lemma 2); (ii) a *geometry-scaled* Jacobi-defect bound with curvature/incidence dependence (Lemma 4); (iii) a DEC consistency result with a frontier term controlled by No-Zeno (Proposition 2).

2.1 Whitney stratifications and tangent structures

Definition 1 (Whitney stratified pseudomanifold)**.** Let G be a second-countable, Hausdorff space with a finite disjoint decomposition $G = \bigsqcup_{i\in I} M_i$ into connected, smooth manifolds (strata), satisfying: (i) the frontier condition; (ii) Whitney (a) and (b) regularity for incident pairs (M_i, M_j) with $M_i \prec M_j$. Each M_i carries a smooth Riemannian metric g_i; the collection is a *stratified Riemannian structure*.

When needed we work in an ambient smooth manifold $\widetilde{G}$ and view strata as embedded submanifolds with induced metric and second fundamental form. We write $TG = \bigsqcup_i TM_i$ and denote the tangent *cone* at $x \in \overline{M_i}$ by $C_x M_i$.

Definition 2 (Frontier-transversal vector fields)**.** A family $V = \{V_i\}_{i\in I}$ with $V_i \in \mathfrak{X}(M_i)$ is *frontier-transversal* if for every incident pair $M_i \prec M_j$ and $x \in M_i \cap \overline{M_j}$ there exist extensions $\widetilde{V}_j$ to a tubular neighborhood of M_i in $\widetilde{G}$ such that $\lim_{y\to x,\, y\in M_j} \mathrm{proj}_{T_x M_i} \widetilde{V}_j(y)$ exists and $\lim_{y\to x,\, y\in M_j} \mathrm{proj}_{N_x M_i} \widetilde{V}_j(y) = 0$.

2.2 Filippov selection and No-Zeno

Trajectories may cross strata. We model dynamics by a differential inclusion $\dot{x} \in F(x)$ on $X = \Gamma \times G$ defined stratumwise and prolonged to frontiers by Filippov regularization. Write $x = (\gamma, g)$.

Definition 3 (Filippov regularization on stratified spaces)**.** Let $F_i : \Gamma \times M_i \rightrightarrows T(\Gamma \times M_i)$ be bounded, measurable with closed, convex values and locally one-sided Lipschitz in x. The Filippov regularization $\mathcal{F}$ associates to each $x \in \Gamma \times G$ the convex closure of essential values of F_j in neighborhoods approaching x along strata with x in their closure. A Carathéodory solution is an absolutely continuous curve $x(\cdot)$ with $\dot{x}(t) \in \mathcal{F}(x(t))$ a.e.

Assumption 1 (Non-explosive frontier dynamics)**.** (i) F_i has linear growth and one-sided Lipschitz modulus L_F on each stratum; (ii) there exists a Lyapunov function whose sublevel sets intersect only finitely many strata.

Remark 1 (Normals on Whitney frontiers and failure set). Choose $n(x)$ as the unit normal field induced by a C^1 tubular neighborhood of the embedded frontier representative in the ambient $\widetilde{G}$. Strict transversality may fail on a set of codimension ≥ 1; the set of *crossing times* that hit this failure set has one-dimensional Lebesgue measure 0.

Lemma 1 (Transversality under C^1 perturbations)**.** *There exists a residual (dense G_δ) set of C^1 perturbations of the stratumwise fields for which $\inf_{x\in\Sigma} \inf_{v\in\mathcal{F}(x)} |\langle v, n(x)\rangle| \geq \alpha > 0$. Equivalently, for a generic perturbation the evaluation map $(x, v) \mapsto \langle v, n(x)\rangle$ has 0 as a regular value on Σ, so tangencies occur only on a set of codimension ≥ 1 and do not affect crossing-time measures.* Uses: *Sard-type argument on $\langle v, n\rangle$; C^1 tubular $n(\cdot)$ from Remark 1.*

Lemma 2 (No-Zeno under strict transversality)**.** [Rep/Alg/Stoch] *Let Σ be a smooth frontier with unit normal n. Suppose one-sided Lipschitz per stratum and strict transversality:*

$$\inf_{x\in\Sigma} \inf_{v\in\mathcal{F}(x)} |\langle v, n(x)\rangle| \;\geq\; \alpha > 0.$$

Here $n(x)$ is as in Remark 1; by Lemma 1 the hypothesis is generic under arbitrarily small C^1 perturbations. Then for any compact interval $[0, T]$, the number of crossings $N_{\mathrm{cross}}(0, T)$ is finite and obeys

$$N_{\mathrm{cross}}(0,T) \;\leq\; \frac{T}{\tau_{\min}}, \qquad \tau_{\min} \;=\; \frac{1}{L_F} \log\Bigl(1 + \frac{L_F\, r_{\mathrm{tube}}}{\alpha}\Bigr) \;\geq\; \frac{r_{\mathrm{tube}}}{\alpha + L_F r_{\mathrm{tube}}}\,.$$

Proof sketch. Within a tubular neighborhood of radius r_{tube} around Σ, let $d(t)$ be the signed normal distance to Σ (positive on one side). Along a Filippov solution, $\dot{d}(t) = \langle \dot{x}(t), n\rangle + \underbrace{\langle x - \pi_\Sigma(x), \nabla n\rangle}_{\text{curvature}}$, with the curvature term bounded by $L_F d(t)$ due to one-sided Lipschitzness and smoothness of n. Strict transversality gives $\dot{d} \geq \alpha - L_F d$, whose comparison solution yields the time to move from $-r_{\mathrm{tube}}$ to $+r_{\mathrm{tube}} \geq \tau_{\min}$; hence at most $T/\tau_{\min}$ crossings on $[0, T]$. □

Uses: (C1) r_{tube}; (C3) L_F and $\tau_{\min}$.

2.3 Bigraded exterior calculus on $T\Gamma \oplus TG$

On a stratum $M_i \subset G$ the product $\Gamma \times M_i$ carries classical multivector calculus. Set $\Lambda^{p,q}(\Gamma, M_i) := \Lambda^p(T\Gamma) \hat{\otimes} \Lambda^q(TM_i)$. Operations (mixed wedge $\wedge$, interior contraction ι, Schouten–Nijenhuis bracket $[\cdot,\cdot]_{SN}$) are standard and understood stratumwise.

2.4 Hybrid Calculus Axioms and geometry-scaled defect

Lemma 3 (Hybrid Calculus Axioms)**.** [Alg] *Let fields/forms be specified stratumwise with frontier-transversal compatibility (Def. 2). Along any Filippov trajectory x_t, the classical identities hold a.e. inside strata: graded-commutativity, contraction antiderivation, and bracket derivation. Jacobi holds up to a frontier defect Δ_{fr} with*

$$\int_0^T \|\Delta_{\mathrm{fr}}(x_t)\|\, dt \;\le\; C_{\mathrm{fr}}(U,T)\, N_{\mathrm{cross}}(0,T),$$

where C_{fr} depends on uniform bounds on fields and their first derivatives near frontiers and one-sided Lipschitz moduli.

Proof sketch. Test Jacobi on a smooth partition subordinate to a frontier tube. The only non-canceling terms are distributional fluxes supported on Σ, controlled by jumps of contractions across Σ; these are bounded by uniform field norms and Lipschitz moduli, yielding the stated L^1 estimate per crossing. □

Uses: (C2) $S_{\max}, G_{\max}$; (C3) one-sided Lipschitz L_F.

Lemma 4 (Jacobi defect with explicit geometry/angle scaling)**.** [Alg] *Under the hypotheses of Lemma 3 with frontier curvature $\kappa_{\max}$, tube radius r_{tube}, and minimal nonzero incidence angle $\theta_{\min}$, the per-crossing defect obeys*

$$\int_0^T \|\Delta_{\mathrm{fr}}(x_t)\|\, dt \;\le\; C_{\mathrm{fr}} \cdot N_{\mathrm{cross}}(0,T), \quad C_{\mathrm{fr}} \;\le\; (\|S\|_\infty + \|\nabla S\|_\infty)\left(1 + c_1\, \kappa_{\max} r_{\mathrm{tube}}\right) \frac{c_2}{\sin\theta_{\min}},$$

with absolute constants c_1, c_2 independent of units, uniformly on compact sublevel sets.

Proof sketch. The tube interaction time scales like $r_{\mathrm{tube}}/\sin\theta$; curvature perturbs n by $O(\kappa_{\max} r_{\mathrm{tube}})$. The distributional commutator across Σ is controlled by jumps in contractions bounded by $\|S\|_\infty + \|\nabla S\|_\infty$, yielding the product form. □

Uses: (C1) $\kappa_{\max}, \theta_{\min}, r_{\mathrm{tube}}$; (C2) $S_{\max}, G_{\max}$; (C3) $\tau_{\min}$.

Proposition 1 (Frontier-compatibility of derivations)**.** [Alg] *If derivations D_i agree in the frontier-transversal sense, their induced actions on $\Lambda^{\bullet,\bullet}$ satisfy the axioms with the same defect bound as in Lemma 3.*

Uses: (C1) frontier tubes; (C2) $S_{\max}, G_{\max}$; (C3) one-sided Lipschitz.

Numerical API (Frontier calculus: crossings, dwell time, geometry-scaled defect)

[Rep/Alg/Comp] This API detects frontier crossings, estimates a uniform dwell time $\widehat{\tau}_{\min}$ and a per-crossing Jacobi-defect budget $\widehat{C}_{\mathrm{fr}}$, and applies the frontier *freeze* rule used later by control and estimation layers.

Inputs. (i) A signed-distance or implicit frontier model $\phi(x)$ on $X = \Gamma \times G$ (positive/negative on adjacent strata); (ii) time-stamped states/velocities $(x_t, \dot{x}_t)$; (iii) tubular radius r_{tube}; (iv) field bounds $S_{\max}, G_{\max}$ and geometric moduli $\kappa_{\max}, \theta_{\min}$ (cf. §); (v) optional discrete mesh for DEC.

Outputs. Crossing log with times and incidence estimates; $\widehat{\tau}_{\min}$; $\widehat{C}_{\mathrm{fr}}$; freeze intervals; optional DEC error audit.

```python
# == 2.1 Frontier tube & normals ==============================================
def build_frontier_tube(phi, r_tube):
    # Returns callable: signed distance d(x), unit normal n(x) on |phi(x)|<=r_tube
    def d(x): return phi(x)                    # assume phi ~ signed dist near frontier
    def n(x):                                  # robust normal from gradient
        g = grad_phi(x); return g / max(norm(g), 1e-12)
    return d, n

# == 2.2 Crossing detection & incidence metrics ================================
def detect_crossings(times, x, d, n, theta_min):
    # Returns list of crossings with normal component v_n and incidence angle
    crossings = []
    for k in range(1, len(times)):
        if d(x[k-1]) * d(x[k]) <= 0:           # sign change (robustify with hysteresis)
            v = (x[k] - x[k-1]) / (times[k]-times[k-1])
            v_n = abs(dot(v, n(x[k])))
            theta = max(theta_min, arccos( min(1.0, max(-1.0, dot(normalize(v), n(x[k]))))  ))
            crossings.append({"t": times[k], "v_n": v_n, "theta": theta})
    return crossings

# == 2.3 Dwell time estimator (No-Zeno proxy) ==================================
def estimate_dwell_time(crossings, horizon_T):
    # use nearest-neighbor spacing between crossings
    if len(crossings) < 2: return {"tau_min_hat": horizon_T, "N_cross": len(crossings)}
    gaps = [crossings[i]["t"] - crossings[i-1]["t"] for i in range(1, len(crossings))]
    tau_hat = max(min(gaps), 0.0)
    return {"tau_min_hat": tau_hat, "N_cross": len(crossings), "gaps": gaps}

# == 2.4 Geometry-scaled defect budget =========================================
def estimate_C_fr(S_max, G_max, kappa_max, theta_vals):
    # heuristic upper bound: curvature + shallow incidence amplify interaction time
    inv_sin = [1.0/max(1e-3, sin(theta)) for theta in theta_vals]   # ~ 1/sin(theta)
    amp = max(inv_sin) if inv_sin else 1.0
    C0 = c_calib * (S_max + G_max) * (1.0 + kappa_max) * amp        # c_calib via unit tests
    return {"C_fr_hat": C0}

# == 2.5 Freeze rule scheduling ================================================
def schedule_freeze(crossings, tau_min_hat):
    # freeze on [t_k - eps, t_k + eps], with eps chosen <= tau_min_hat/2
    eps = 0.5 * tau_min_hat
    return [{"t0": c["t"] - eps, "t1": c["t"] + eps} for c in crossings]

# == 2.6 Optional DEC audit ====================================================
def dec_consistency(mesh_h, k_order, N_cross):
    # stratumwise O(h^k) + frontier O(h^k) per crossing
    err = C_dec * (mesh_h**k_order) * (1 + N_cross)
    return {"dec_err_bound": err, "k": k_order, "h": mesh_h}

# == 2.7 Orchestration =========================================================
def frontier_runtime(times, x, phi, r_tube, S_max, G_max, kappa_max, theta_min, mesh=None):
    d, n = build_frontier_tube(phi, r_tube)
    crossings = detect_crossings(times, x, d, n, theta_min)
    dwell = estimate_dwell_time(crossings, horizon_T=times[-1]-times[0])
    Cfr = estimate_C_fr(S_max, G_max, kappa_max, [c["theta"] for c in crossings])
    freezes = schedule_freeze(crossings, dwell["tau_min_hat"])
    audit = dec_consistency(mesh.h, mesh.k, dwell["N_cross"]) if mesh else {}
```

```
    log = {"N_cross": dwell["N_cross"], "tau_min_hat": dwell["tau_min_hat"],
           "C_fr_hat": Cfr["C_fr_hat"], "freezes": freezes, **audit}
    return log
```

Acceptance checks. (i) Logs report N_{cross}, $\widehat{\tau}_{\min}$, and $\widehat{C}_{\mathrm{fr}}$; (ii) freeze intervals are non-overlapping and contained in the tube; (iii) optional DEC audit prints $O(h^k) + N_{\mathrm{cross}} O(h^k)$.

Uses: r_{tube}, L_{F}, $\theta_{\min}$, $\kappa_{\max}$, $S_{\max}$, $G_{\max}$.

2.5 DEC interoperation with frontier term

Let $\mathcal{T}_h$ be a shape-regular simplicial (or cubical) mesh on each M_i with mesh size h, and let W_h denote Whitney forms. Denote by $\mathsf{W}_h : \Omega^\bullet \to \Omega_h^\bullet$ the Whitney projection and by R_h its de Rham inverse (integration on chains). Assume crossings satisfy Lemma 2 and fields are C^{k+1} on strata.

Proposition 2 (DEC consistency with frontier term and cut-cell quadrature)**.** [Alg/Comp] *Assume either* (a) *fitted meshes with uniform angle bounds on cut cells, or* (b) *unfitted meshes with a ghost-penalty/aggregate stabilization on the tube. For* $0 \le r \le k$ *and each stratum* M_i*,*

$$\|\mathsf{W}_h(\omega \wedge \eta) - \mathsf{W}_h\omega \wedge_h \mathsf{W}_h\eta\|_{L^2(M_i)} \le C\, h^{k-r} \|\omega\|_{H^k} \|\eta\|_{H^k},$$

and similarly for brackets. Along any Filippov trajectory x_t *on* $[0,T]$*,*

$$\int_0^T \|\Delta_{\mathrm{fr},h}(x_t)\|\, dt \;\le\; C\, h^{k-r}\, T \;+\; \underbrace{C_{\mathrm{quad}}\, h^q\, T}_{\text{cut-cell quadrature, } q \ge k} \;+\; C_{\mathrm{fr}}\, N_{\mathrm{cross}}(0,T).$$

Thus the total defect is $O(h^{k-r}T + h^q T) + O(C_{\mathrm{fr}}\, T/\tau_{\min})$.

Uses: Shape regularity or ghost-penalty on cut cells; Lem. 2; Lem. 4.

Acceptance checks for Section 2.

1. All later bounds that invoke frontier disturbance (e.g., ISS in §6) reference both $\tau_{\min}$ (Lemma 2) and C_{fr} (Lemma 4).
2. Any occurrence of N_{cross} appears only inside expressions dominated by $T/\tau_{\min}$ on finite horizons.
3. The DEC bridge exposes explicit h-rates and a frontier contribution that scales as $C_{\mathrm{fr}}\, T/\tau_{\min}$.

3 Strategic Series and Cross-Cumulants

We formalize strategic fields on Γ and G, define scalar evaluation functionals, and construct mixed cumulants under invariant or windowed laws. New in this draft: (i) measurability and L^2 conditions for $T_{u,v}$ (Lemma 5); (ii) a corrected factorization statement that separates the *global* "separating dictionary" case from the *finite-K* practical case (Proposition 3); (iii) a quantitative windowed stability bound (Lemma 6).

3.1 Strategic series and evaluation functionals

On each stratum $M_i \subset G$, take the bigraded series $\hat{S} = \sum_{p,q} S^{(p,q)}$ with

$$S^{(p,q)} \in \Gamma\big(\Lambda^p(T\Gamma) \hat{\otimes} \Lambda^q(TM_i)\big).$$

For covectors $u \in \Lambda^p(T^*\Gamma)$ and $v \in \Lambda^q(T^*G)$ define the scalar *evaluation functional*

$$T_{u,v}(x) \;:=\; \iota_{u\otimes v}\, \hat{S}(x)\,, \qquad x = (\gamma, g) \in \Gamma \times M_i,$$

interpreted stratumwise (Lemma 3 ensures algebraic well-posedness a.e. along Filippov trajectories).

3.2 Existence of laws and windowing

Under Assumption 1, standard Lyapunov/tightness arguments deliver invariant laws μ for $(X_t)_{t\geq 0}$ when the drift/diffusion contracts on average; otherwise we use *windowed laws* on $[t-\tau, t]$:

$$\mu_{[t-\tau,t]} \;:=\; \frac{1}{\tau}\int_{t-\tau}^{t} \delta_{X_s}\, ds.$$

All statements below hold either for μ or uniformly in t for $\mu_{[t-\tau,t]}$ when τ exceeds the local mixing time.

Lemma 5 (Measurability and L^2 of evaluation functionals)**.** [Stoch/Alg] *Suppose on each stratum M_i: (i) $S^{(p,q)}$ is Borel and locally square-integrable; (ii) (u, v) have at most polynomial growth in charts; (iii) crossings satisfy Lemma 2. Then $T_{u,v}$ is Borel on $X = \Gamma \times G$ and $T_{u,v} \in L^2(\mu)$ (or $L^2(\mu_{[t-\tau,t]})$).*

Proof sketch. Carathéodory measurability of contractions in local charts and the frontier-transversal compatibility (Def. 2) yield Borel measurability; local L^2 on strata and finite crossing count ensure global L^2 under μ or window averages. □

Uses: (C2) $S_{\max}$ bounds; (C3) No-Zeno $\tau_{\min}$; Rep charts.

3.3 Mixed cumulants and factorization

Let $\mathcal{T}_K = \{T^{(m)}_{p,q}\}_{1\leq p+q\leq K}$ be a centered, whitened finite dictionary (Section 5). For a finite list $(Z_1, \ldots, Z_m)$ drawn from $\mathcal{T}_K$, define the (alternating) joint cumulant $\kappa(Z_1, \ldots, Z_m)$ via the usual moment–cumulant Möbius inversion; we write $\kappa_{p,q}$ for degree (p, q) cumulants.

Proposition 3 (Factorization via vanishing mixed cumulants)**.** [Stoch/Inf] *Consider γ-only statistics and G-only statistics drawn from (possibly infinite) dictionaries.*

(i) Global, separating dictionary. *If the dictionary separates points and the cumulant generating function is analytic near 0, then*

$$\gamma \;\perp\; G \;\big|\; Y \qquad\Longleftrightarrow\qquad \kappa_{p,q} = 0 \;\; \textit{for all } p, q \geq 1.$$

(ii) Finite-K implication with quantitative residual. *Fix total order $K \in \{2, 3, \ldots\}$ and a centered, whitened finite dictionary $\mathcal{T}_K$. If $\kappa_{p,q} = 0$ for all $p, q \geq 1$ with $p + q \leq K$, then for any polynomial functional F of $\mathcal{T}_K$ with total degree $\leq K$,*

$$\Big|\,\mathbb{E}[F(\gamma, G)] \;-\; \mathbb{E}[F_\Gamma(\gamma)]\,\mathbb{E}[F_G(G)]\,\Big| \;\leq\; C_K(F) \sum_{p+q\geq K+1} \|\kappa_{p,q}\|,$$

where $C_K(F)$ depends only on K and the coefficients of F in the whitened basis. In particular, the log-moment generating function satisfies

$$\psi(\theta_\Gamma, \theta_G) \;=\; \psi_\Gamma(\theta_\Gamma) + \psi_G(\theta_G) \;+\; R_{K+1}(\theta), \qquad |R_{K+1}(\theta)| \;\leq\; C'_K \sum_{p+q\geq K+1} \|\kappa_{p,q}\|\,\|\theta\|^{p+q},$$

for $\|\theta\|$ within the EF radius of Section 5.

Proof sketch. (i) follows from the Lévy–Khintchine/moment–cumulant expansion with a separating dictionary. (ii) truncates the series at total degree K; the remainder aggregates neglected mixed cumulants with combinatorial coefficients (Bell polynomials) that are bounded by $C_K(F)$ after whitening. The EF radius control comes from Theorem 2/Remark **??**. □

Uses: (C5) whitened $F = \mathrm{FIM}(0) \succ 0$, EF radius $r \leq \epsilon/L_\psi$.

Interpretation (cross-terms). Cross-terms in $\hat{S}$ map to *mixed* cumulants $\kappa_{p,q}$: nonzero $\kappa_{p,q}$ for $p, q \geq 1$ are the irreducible coupling certificates between decision flow on Γ and structure updates on G. Proposition 3 formalizes the "no cross-term ⇒ factorization up to order K" claim used in closure policies.

3.4 Windowed stability under drift

Let L be the W_1 drift speed of the ground-truth law across time (Section 4). For windowed cumulants computed on $[t-\tau, t]$ and $[s-\tau, s]$ with $|t-s| \ll \tau$:

Lemma 6 (Windowed stability of mixed cumulants)**.** [Stoch] *If each statistic in $\mathcal{T}_K$ is L_T-Lipschitz in d_X, then for $p+q \le K$,*

$$\left|\kappa_{p,q}^{(t,\tau)} - \kappa_{p,q}^{(s,\tau)}\right| \;\le\; C_{p,q}\, L_T\, \frac{|t-s|}{\tau} \;+\; o_\tau(1),$$

uniformly on compact sublevel sets. The constant $C_{p,q}$ depends only on p, q and the dictionary normalization.

Proof sketch. Write cumulants as multilinear polynomials in moments; bound each moment difference by the Kantorovich–Rubinstein dual with L_T and the window averaging factor $|t-s|/\tau$, then propagate through the multilinear map. □

Uses: (C4) τ; Lipschitz statistics; window laws $\mu_{[t-\tau,t]}$.

Acceptance checks for Section 3.

1. Proposition 3 *(ii)* is the only statement invoked when working with a finite dictionary; any "⇔" claim elsewhere references *(i)* with a separating dictionary.
2. All L^2 and measurability claims for $T_{u,v}$ cite Lemma 5.
3. Any time-variation bounds for windowed cumulants cite Lemma 6 with explicit dependence on $|t-s|/\tau$.

Numerical API (Strategic series, windowed cumulants, finite-K tests)

[Stoch/Inf/Comp] This API constructs test functionals $T_{u,v}$, performs invariant whitening, computes windowed cumulants up to order $K \le 3$ by streaming U-statistics, and runs a finite-K factorization test (â‡', not â‡").

Inputs. (i) Centered raw features/features-of-forms for letters $a_{p,q}^{(m)}$; (ii) symmetry group actions (frames on Γ, graph isomorphisms on G) for invariant whitening; (iii) window spec (τ, stride); (iv) target order $K \in \{2, 3\}$.

Outputs. Per-window $\widehat{\mathbf{k}}_K$, FIM norm, HAC/BB hooks (from Section 9), and a BH/BY-controlled finite-K factorization report.

```
# == 3.1 Invariant whitening ======================================================
def orbit_invariant_cov(features, group_actions, nsamp=128):
    # Monte Carlo over group actions to enforce symmetry-invariant inner product
    Sigma = 0
    for g in sample(group_actions, nsamp):
        Xg = apply_action(features, g)           # permute/rotate letters
        Sigma += cov(Xg)
    Sigma /= nsamp
    return Sigma

def whiten_dictionary(features, Sigma):
    W = inv(cholesky(Sigma))                    # Fisher/whitening map
    return features @ W.T, W                    # whitened letters, map

# == 3.2 Test functionals T_{u,v} ================================================
def build_T_uv(letters, u, v):
    # u,v are contraction descriptors into bigraded series \hat S
```

```
    def T(x):
        # evaluate contractions using current \hat S and x=(\gamma,G)
        return contract(u, v, letters, x)
    return T

# == 3.3 Streaming cumulants (order <= 3) ===================================
def stream_cumulants(T_stream, K=3):
    # maintains running moments and converts to cumulants via M\"obius inversion
    state = init_u_stats(K)
    for Tt in T_stream: update_u_stats(state, Tt, K)
    return u_to_cumulants(state, K)             # dict of kappa_{p,q}^{(m)}

# == 3.4 Windowed pipeline ==================================================
def windowed_cumulants(data_stream, letters, Sigma, K, win_size, stride):
    T_wh, W = whiten_dictionary(letters, Sigma)
    windows = sliding_windows(data_stream, win_size, stride)
    out = []
    for w in windows:
        kappa = stream_cumulants(eval_T_stream(T_wh, w), K=K)
        kvec  = flatten_kappa(kappa)            # stack by (p,q,m)
        F     = fisher_at_P0(T_wh)              # = I if fully whitened
        fim_norm = sqrt(kvec.T @ inv(F) @ kvec)
        out.append({"window": w.id, "kappa": kappa, "kvec": kvec, "fim_norm": fim_norm})
    return out

# == 3.5 Finite-K factorization test (BH/BY) ================================
def finiteK_factorization_test(window_out, groups, q=0.1, method="BH"):
    # H0: selected entries of mixed cumulants == 0  (vanish up to K)
    pvals = {}
    for win in window_out:
        pvals[win["window"]] = test_zero_kappa(win["kvec"])  # HAC/BB from section 9
    # group by bidegree/stratum per section 9
    rejections = {}
    for gid, P in groups.items():
        rejections[gid] = fdr_bh(P, q) if method=="BH" else fdr_by(P, q)
    return {"pvals": pvals, "reject": rejections}

# == 3.6 Diagnostics ========================================================
def mixing_diagnostic(series, max_lag=64):
    ac = autocorr(series, max_lag)
    return {"ac_tail": ac[-5:], "mix_flag": (abs(ac[max_lag])<0.1)}
```

Acceptance checks. (i) Whitening uses orbit-averaged Σ (equivariant); (ii) each window reports $\|\widehat{\mathbf{k}}_K\|_{\mathrm{FIM}^{-1}}$; (iii) finite-$K$ test declares only the $\Rightarrow$ direction (vanishing up to K $\Rightarrow$ factorization up to K), with BH/BY and group structure as in Section 9; (iv) mixing diagnostic is logged for each window.

Uses: Invariant whitening (Assumption 3); Fisher F; Section 9 HAC/BB/FDR.

4 Hierarchical Closure: BBGKY Analogy and DRO

We pose closure as a Wasserstein–KL distributionally robust optimization (DRO) and derive a tractable soft-transport dual. New here: (i) a stratified *reference measure* λ that is *refinement invariant* (Definition 4, Proposition **??**); (ii) measurability and stratum-Lipschitz properties of the soft potential φ_η (Lemma 8); (iii) differentiability of $\mathcal{R}_\rho$ in (θ, η) with an envelope/KKT system and explicit gradients (Proposition 5); (iv) an *adaptive radius* policy with guarantees that expose finite-sample W_1 rates via an effective transport dimension d_{eff} (Definition 5, Theorem 1).

4.1 DRO objective and soft-transport dual

Given a whitened statistic dictionary up to order K (Section 5), we fit θ by

$$\min_{\theta\in\Theta} \underbrace{\sup_{Q:\ W_1(Q,\hat{P})\le\rho} D_{\mathrm{KL}}(Q\|P_\theta)}_{:=\ \mathcal{R}_\rho(\theta)} + \lambda_{\mathrm{hopf}} \left\|(\mathrm{id}-\Pi_{\le K})\,\hat{S}\right\|^2. \tag{1}$$

Here W_1 is computed with a ground metric d_X on $X=\Gamma\times G$, and the second term penalizes Hopf-residual energy (Section 8).

Definition 4 (Stratified reference with continuous+atomic split)**.** Write $\lambda=\lambda_{\mathrm{cont}}+\lambda_{\mathrm{atom}}$ with

$$\lambda_{\mathrm{cont}} := \sum_{i\in I} w_i\,\mathcal{H}^{\dim M_i}\lfloor_{M_i}, \qquad \lambda_{\mathrm{atom}} := \sum_{j\in J} w'_j\,\delta_{D_j},$$

normalized on compacta and refinement-invariant (Prop. 4). For any model P_θ, write $P_\theta = P_\theta^{\mathrm{cont}}+P_\theta^{\mathrm{atom}}$ with $P_\theta^{\mathrm{cont}}\ll\lambda_{\mathrm{cont}}$ and $P_\theta^{\mathrm{atom}}\ll\lambda_{\mathrm{atom}}$.

Proposition 4 (Refinement-invariant stratified reference measure)**.** [Stoch/Clos] *Let $G=\bigsqcup_i M_i$ be Whitney-stratified. Choose weights $c_i\propto\mathcal{H}^{\dim M_i}(M_i\cap K)$ on compacta K (Hausdorff volume of top-dimensional pieces restricted to K) and assign atom weights c'_j continuously under refinements so that $\sum_i c_i+\sum_j c'_j=1$ on K. If two stratifications admit a common refinement, the induced measures λ are mutually absolutely continuous and the inner value $\mathcal{R}_\rho(\theta)$ is identical for both.*

Proof sketch. Use Carathéodory's construction on the stratified σ-algebra and note that Hausdorff measures agree up to equivalence across common refinements on each piece; atomic mass continuity prevents mass creation under partitioning. The dual integral defining φ_η is invariant under equivalent λ. □

Uses: Whitney stratification; Hausdorff measures; Carathéodory extension.

Lemma 7 (Differentiability and KKT envelope)**.** [Stoch/Clos/Inf] *Assume: (i) $dP_\theta/d\lambda$ is C^1 in θ on each $\Gamma\times M_i$; (ii) there exists an $L^1(\lambda)$ envelope G with $\sup_{\|\theta\|\le r,\ \eta\in[0,\eta_{\max}]}\left\|\partial_\theta\log\frac{dP_\theta}{d\lambda}\right\|\le G$; and (iii) d_X is continuous. Then*

$$\nabla_\theta\mathcal{R}_\rho(\theta) \;=\; \nabla_\theta\mathbb{E}_{\hat{P}}[\varphi_{\eta^\star}(\cdot;\theta)] \;=\; \mathbb{E}_{\hat{P}}\left[\frac{\int e^{-\eta^\star d_X(z,\cdot)}\,\partial_\theta\frac{dP_\theta}{d\lambda}(z)\,d\lambda(z)}{\int e^{-\eta^\star d_X(z,\cdot)}\,\frac{dP_\theta}{d\lambda}(z)\,d\lambda(z)}\right],$$

and the KKT system holds with $\eta^\star\ge 0$ solving $\rho=\mathbb{E}_{\hat{P}}[\partial_\eta\varphi_\eta]$.

Proof sketch. Carathéodory measurability (Lemma 8) + dominated convergence (envelope) justify interchanging ∇_θ and the integral; apply the envelope theorem for the inner infimum in η. □

Uses: Envelope $G\in L^1$; Lem. 8.

Remark 2 (Finite-sample W_1 drift rates and δ in Thm. 1). *[Stoch/Clos]* With n points per window and effective transport dimension d_{eff}, $|\hat{L}_t-L|=O_{\mathbb{P}}\big(n^{-1/d_{\mathrm{eff}}}\big)$ in the worst case, and improves to $O_{\mathbb{P}}(n^{-1/2})$ under bounded transport dimension / sub-Gaussian dictionary tails. Choose δ in Theorem 1 accordingly to set (c,τ) pragmatically.

Uses: Window size n; dictionary tails/transport-dimension assumption.

With $P_\theta\ll\lambda$ and continuous d_X, the inner sup admits the Donsker–Varadhan/Kantorovich dual

$$\mathcal{R}_\rho(\theta) \;=\; \inf_{\eta\ge 0}\Big\{\eta\rho+\mathbb{E}_{\hat{P}}[\varphi_\eta(\cdot;\theta)]\Big\}, \quad \varphi_\eta(x;\theta) \;:=\; \log\int\exp\big(-\eta d_X(z,x)\big)\,\frac{dP_\theta}{d\lambda}(z)\,d\lambda(z).$$

Lemma 8 (Measurability and stratum-Lipschitz of φ_η)**.** [Stoch/Clos] *If d_X is continuous and $z\mapsto\frac{dP_\theta}{d\lambda}(z)$ is stratumwise continuous, then $x\mapsto\varphi_\eta(x;\theta)$ is Borel on X and locally Lipschitz on each $\Gamma\times M_i$.*

Proof sketch. Carathéodory's criterion applies to $(x,z)\mapsto e^{-\eta d_X(z,x)}\,dP_\theta/d\lambda(z)$; local dominated convergence on strata yields stratum-Lipschitzness of φ_η. □

Uses: (C1) stratum charts; continuity of d_X; stratumwise continuity of $dP_\theta/d\lambda$.

4.2 Differentiability and KKT system

Proposition 5 (Differentiability and KKT for the soft-transport dual)**.** [Clos/Inf] *Suppose for some $r > 0$, $\eta \in [0, \eta_{\max}]$, and all $\|\theta\| \le r$: (i) $dP_\theta/d\lambda$ is C^1 stratumwise with an $L^1(\lambda)$ envelope uniform in (θ, η); (ii) $\varphi_\eta(x;\theta)$ from Lemma 8 is $\hat{P}$-integrable. Then $\mathcal{R}_\rho(\theta)$ is C^1 on $\{\|\theta\| \le r\}$, and the inner minimization over $\eta \ge 0$ admits KKT conditions with*

$$\partial_\eta\Big(\eta\rho + \mathbb{E}_{\hat{P}}[\varphi_\eta]\Big)\Big|_{\eta=\eta^\star} \;=\; 0 \quad\Longleftrightarrow\quad \rho \;=\; \mathbb{E}_{\hat{P}}\Big[\underbrace{\mathbb{E}_{Z\sim Q^x_{\eta^\star,\theta}}\big[d_X(Z,x)\big]}_{:=\,\bar{d}_\eta(x;\theta)}\Big],$$

where $Q^x_{\eta,\theta}$ is the Gibbs kernel on X with density

$$\frac{dQ^x_{\eta,\theta}}{d\lambda}(z) \;\propto\; e^{-\eta d_X(z,x)}\,\frac{dP_\theta}{d\lambda}(z).$$

Moreover,

$$\nabla_\theta \mathcal{R}_\rho(\theta) \;=\; \mathbb{E}_{\hat{P}}\Big[\;\mathbb{E}_{Z\sim Q^x_{\eta^\star,\theta}}\big[\;\nabla_\theta \log(dP_\theta/d\lambda)(Z)\;\big]\;\Big].$$

If $\eta \mapsto \mathbb{E}_{\hat{P}}[\varphi_\eta]$ is strictly convex (e.g. when d_X is non-degenerate on supports), then $\eta^\star$ is unique and the envelope theorem applies.

Proof sketch. Differentiate φ_η under the integral using the uniform L^1 envelope; apply Danskin's theorem/envelope theorem at the η-minimizer. The ∂_η identity is the well-known soft-transport moment. The θ-gradient follows from the log-derivative trick inside the Gibbs kernel. □

Uses: Dominated convergence envelope; (C5) EF radius context; (C4) window integrability.

Corollary 1 (KKT and gradient under $\lambda_{\text{cont}} + \lambda_{\text{atom}}$)**.** *With Definition 4, the soft-transport dual reads $\mathcal{R}_\rho(\theta) = \inf_{\eta\ge 0}\{\eta\rho + \mathbb{E}_{\hat{P}}[\varphi_\eta]\}$ with*

$$\varphi_\eta(x;\theta) = \log\Big(\int e^{-\eta d_X(z,x)}\,\frac{dP^{\text{cont}}_\theta}{d\lambda_{\text{cont}}}(z)\,d\lambda_{\text{cont}}(z) \;+\; \sum_{D\in\text{Atoms}} e^{-\eta d_X(D,x)}\,\frac{dP^{\text{atom}}_\theta}{d\lambda_{\text{atom}}}(D)\Big).$$

The KKT condition is $\rho = \mathbb{E}_{\hat{P}}[\bar{d}_\eta(x;\theta)]$ with $\bar{d}_\eta$ the $Q^x_{\eta,\theta}$-expectation over the product *measure, and*

$$\nabla_\theta \mathcal{R}_\rho(\theta) = \mathbb{E}_{\hat{P}}\Big[\mathbb{E}_{Z\sim Q^x_{\eta^\star,\theta}}\big[\nabla_\theta \log(dP^{\text{cont}}_\theta/d\lambda_{\text{cont}})(Z)\big]\Big] \;+\; \sum_D \hat{P}(\{x\})\,\pi_{\eta^\star,\theta}(D\mid x)\,\nabla_\theta \log\frac{dP^{\text{atom}}_\theta}{d\lambda_{\text{atom}}}(D).$$

Lemma 9 (Uniqueness of $\eta^\star$ by strict decrease)**.** *If $d_X(\cdot, x)$ is not λ-a.e. constant for each x (GKYP-type nondegeneracy), then for fixed (θ, x)*

$$g'_x(\eta) = -\operatorname{Var}_{Q^x_{\eta,\theta}}\big[d_X(Z,x)\big] \;<\; 0.$$

Hence $g(\eta) := \mathbb{E}_{\hat{P}}[g_x(\eta)]$ is strictly decreasing and the KKT root $\eta^\star$ is unique.

Proposition 6 (DRO-metric sensitivity)**.** *Let $d^\epsilon_X = d_X + \epsilon\,\delta d$ with δd Lipschitz and $\theta^*(\epsilon)$ be the minimizer of (1). Under the IFT assumptions for the outer problem, $\partial_\epsilon\theta^*(0) = -\big[\nabla^2_\theta \mathcal{R}_\rho(\theta^*)\big]^{-1}\nabla_\theta \mathbb{E}_{\hat{P}}[\partial_\epsilon\varphi_{\eta^*}(\cdot;\theta^*)]$, with $\partial_\epsilon\varphi_{\eta^*}(x;\theta) = -\eta^*\,\mathbb{E}_{Q^x_{\eta^*,\theta}}[\delta d(Z,x)]$.* Report: *log $\|\partial_\epsilon\theta^*(0)\|$ as "DRO-metric sensitivity".*

Numerical note. Compute φ_η with log-sum-exp stabilization; sample from $Q^x_{\eta^\star,\theta}$ via importance sampling or short-run MCMC confined to a stratum (accept/reject at frontiers).

4.3 Adaptive radius policy: finite-sample rates

Definition 5 (Adaptive radius policy)**.** *[Clos/Stoch]* For window size τ with n effective observations and transport dimension d_{eff} for (X, d_X) on the window, define a pilot drift estimate

$$\hat{L}_t \;:=\; \tfrac{1}{\tau}\, W_1\big(\hat{P}_{[t-\tau,t]}, \hat{P}_{[t-2\tau,t-\tau]}\big), \qquad \rho_t \;:=\; c\,\hat{L}_t\,\tau \quad (c \geq 1).$$

Uses: (C4) τ, ρ; (C1) metric measurability.

Theorem 1 (Containment and regret under finite-sample W_1 rates)**.** [Clos] *Let $P_{[t-\tau,t]}$ and $P_{[t-2\tau,t-\tau]}$ be the true window laws with W_1 drift speed L. Suppose $\hat{P}$'s are empirical with n effective points and*

$$|\hat{L}_t - L| \;\leq\; \delta_{n,d_{\text{eff}}} \quad \textit{with} \quad \delta_{n,d_{\text{eff}}} = \begin{cases} C\, n^{-1/d_{\text{eff}}} & \textit{(worst-case, no moment/low-dim assumptions)}, \\ C'\, n^{-1/2} & \textit{(under bounded transport dimension / sub-Gaussian tails)}. \end{cases}$$

Then with $\rho_t = c\,\hat{L}_t\,\tau$:

1. *(*Containment*) $P_{[t-\tau,t]} \in \mathcal{W}_1(\hat{P}_{[t-\tau,t]}, \rho_t)$ with probability $\geq 1-\alpha$ for a choice of $c = c(\alpha)$ depending on the tail bound of $\delta_{n,d_{\text{eff}}}$.*
2. *(*Regret drift term*) The drift contribution in $D_{\mathrm{KL}}(P\|P_{\widehat{\theta}})$ scales as*

$$\text{Regret}_{\text{drift}} \;\leq\; C_d\big(L + \delta_{n,d_{\text{eff}}}\big)\,\tau.$$

Proof sketch. Triangle inequality for W_1 and concentration of empirical W_1 around truth give the containment for appropriate c. The regret bound follows from the dual representation with Lemma 8 (Lipschitz in x) and Proposition 5 to control the variation of $\varphi_{\eta^\star}$ across adjacent windows. □

Uses: (C4) τ; Lipschitz constants from Lemma 8; envelope from Proposition 5.

4.4 Error decomposition (updated)

With the adaptive radius, the standard decomposition becomes

$$D_{\mathrm{KL}}(P\|P_{\widehat{\theta}}) \;\leq\; C_a\big\|(\mathrm{id} - \Pi_{\leq K})\hat{S}\big\|^2 \;+\; C_e n^{-1/2} \;+\; C_d\big(L + \delta_{n,d_{\text{eff}}}\big)\,\tau,$$

where the last term replaces the purely qualitative drift piece and is reported per window in logs together with $\eta^\star$.

Acceptance checks for Section 4.

1. Every use of the soft-transport dual references Definition 4 and Lemma 8; claims about invariance under re-stratification cite Proposition **??**.
2. KKT/gradient statements cite Proposition 5 and list the envelope hypotheses; implementations log $\eta^\star$.
3. The adaptive radius and regret bounds cite Theorem 1 with explicit n and d_{eff} dependence; runs log $\widehat{L}_t$, ρ_t, $\eta^\star$.

Numerical API (soft-transport dual, gradients, and logging)

```
# Ground types (conceptual)
Metric dX(x,z)                    # ground metric on X = Gamma x G
Model  P_theta                    # exponential-family or other density; exposes:
                                  #   logpdf(z; theta), score(z; theta) = grad_theta logpdf
```

```
Measure lambda                        # refinement-invariant reference (Def.~4)
Sampler draw_from_lambda(stratum, m)  # draws z~lambda restricted to a stratum (or global)
Projector Pi_leqK                     # Hopf projector used in residual penalty
Logger log_json(dict)                 # appends a JSON line to the run log

# ---------- Core primitives (stable numerics) ----------
function varphi_eta(x_batch, theta, eta, dX, P_theta, lambda, strata_policy):
    # returns: vector phi[i] = varphi_eta(x_i; theta)
    for each x in x_batch:
        S = 0.0            # log-sum-exp accumulator (log domain)
        w_sum = 0.0
        # importance base: sample M support points from lambda (same stratum as x by default)
        Z = draw_from_lambda(strata_policy(x), M)
        # compute log-weights with log-sum-exp stabilization
        for z in Z:
            \ell = -eta * dX(z, x) + logpdf_P_over_lambda(z; theta, P_theta, lambda)
            # standard log-sum-exp trick
            if S == 0.0: S = \ell
            else:
                if \ell > S: w_sum = exp(S - \ell) * w_sum + 1.0; S = \ell
                else:        w_sum = w_sum + exp(\ell - S)
        phi[x] = S + log(w_sum) - log(M)        # Monte Carlo approx. to log \int ...
    return phi

function soft_transport_kernel(x, theta, eta, dX, P_theta, lambda, strata_policy):
    # returns a closure that can sample/reweight from Q_{eta,theta}^x via importance weights on lambda
    Z = draw_from_lambda(strata_policy(x), M)
    L = [ -eta*dX(z,x) + logpdf_P_over_lambda(z; theta, P_theta, lambda) for z in Z ]
    S = max(L); w = [ exp(\ell - S) for \ell in L ]; w = w / sum(w)   # normalized weights
    return (Z, w)   # usage: E_{Q}[f] \approx sum_j w[j] * f(Z[j])

# ---------- Inner solver for \eta^\star ----------
function eta_star(rho, x_batch, theta, dX, P_theta, lambda, strata_policy,
                  bracket=[1e-6, 1e3], tol=1e-5, M=512):
    # solve g(eta) = E_x E_{Q_{eta,theta}^x}[ dX(Z,x) ] - rho = 0
    function g(eta):
        acc = 0.0
        for x in x_batch:
            (Z,w) = soft_transport_kernel(x, theta, eta, dX, P_theta, lambda, strata_policy)
            acc += sum_j w[j] * dX(Z[j], x)
        return acc / len(x_batch) - rho
    # bracket monotone root (g is decreasing in eta under mild conditions)
    a,b = bracket
    if g(a) < 0: return a
    if g(b) > 0: increase b geometrically until g(b) < 0 or b > 1e6
    # Brent or bisection (robust)
    while |b-a| > tol*(1+min(|a|,|b|)):
        c = 0.5*(a+b)
        if g(c) > 0: a = c else: b = c
    return 0.5*(a+b)

# ---------- Risk and gradient ----------
function R_rho(theta, x_batch, rho, dX, P_theta, lambda, strata_policy, M=1024):
    eta = eta_star(rho, x_batch, theta, dX, P_theta, lambda, strata_policy, M=M/2)
    phi = varphi_eta(x_batch, theta, eta, dX, P_theta, lambda, strata_policy)
    return eta*rho + mean(phi), eta

function grad_R_rho(theta, x_batch, eta, dX, P_theta, lambda, strata_policy, M=1024):
    # \nabla_\theta R_rho(\theta) = E_x E_{Q_{eta,theta}^x}[ score(Z; theta) ]
```

```
    g = 0 (shape like theta)
    for x in x_batch:
        (Z,w) = soft_transport_kernel(x, theta, eta, dX, P_theta, lambda, strata_policy)
        g += sum_j w[j] * score(Z[j]; theta)
    return g / len(x_batch)

# ---------- Adaptive radius and drift logging ----------
function adaptive_radius(window_curr, window_prev, c, dX, effective_n, d_eff_hint=None):
    # window_curr/prev: datasets; c >= 1
    \hat W1 = empirical_W1(window_curr, window_prev, dX)      # same metric used in dual
    \hat L  = \hat W1 / window_length_tau
    rho_t   = c * \hat L * window_length_tau
    # optional: rate hint for monitoring (not used in computation)
    if d_eff_hint is None: rate = C / sqrt(effective_n) else: rate = C / (effective_n)^(1/d_eff_hint)
    return rho_t, \hat L, \hat W1, rate

# ---------- One-call convenience wrapper per window ----------
function dro_step(theta, window_curr, window_prev, config):
    # 1) adaptive radius (logs drift)
    rho_t, \hat L, \hat W1, rate_hint = adaptive_radius(window_curr, window_prev,
                            config.c_radius, dX=config.dX, effective_n=config.n_eff,
                            d_eff_hint=config.d_eff)
    # 2) pick minibatch x_batch from window_curr (or full window)
    x_batch = minibatch(window_curr, size=config.batch)
    # 3) compute risk and eta*
    R, eta = R_rho(theta, x_batch, rho_t, config.dX, config.P_theta, config.lambda, config.strata_policy, M=config.
    # 4) gradient (for optimizer)
    g = grad_R_rho(theta, x_batch, eta, config.dX, config.P_theta, config.lambda, config.strata_policy, M=config.M_
    # 5) Hopf residual (for penalty term and audit)
    S_resid = norm( (Id - Pi_leqK) @ hatS(window_curr) )^2
    # 6) structured logging
    log_json({
        "window_t": config.t_id,
        "rho_t": rho_t,
        "L_hat": \hat L,
        "W1_hat": \hat W1,
        "eta_star": eta,
        "R_rho": R,
        "hopf_residual": S_resid,
        "n_eff": config.n_eff,
        "rate_hint": rate_hint,
        "M_mc": config.M_mc,
        "batch": config.batch
    })
    return R + config.lambda_hopf * S_resid, g
```

Recommended defaults and guards.

- *Bracketing for $\eta^\star$.* Start with $[10^{-6}, 10^3]$; expand the right endpoint by $\times 10$ if $g(b) > 0$. Enforce a hard cap (e.g., 10^8) and fall back to $\eta^\star = b$ if the bracket fails—log this.
- *Monte Carlo budget.* Use $M = 512$–2048 per x; reuse samples across nearby η during the 1-D solve (common-random-numbers variance reduction).
- *Stratum policy.* By default, draw Z from the *same* stratum as x; allow "cross-stratum" proposals only if the frontier tube flag is raised (then freeze Π and θ as in Section 6/Section 7).
- *Stability.* All integrals in φ_η and kernel weights use log-sum-exp. Clip η to avoid $\exp(-\eta d)$ underflow; report the clip in the log.

- *EF models.* For exponential families, `score(z;theta)` is available in closed form; cache $F = \mathrm{FIM}(0)$ for whitening and preconditioning. Use a Fisher preconditioner for the optimizer step.
- *Unit tests.* (i) At $\rho = 0$, verify $\eta^\star = 0$, $R_\rho(\theta) = \mathbb{E}[\log \frac{dP_\theta}{d\lambda}(X)]$. (ii) If $d_X \equiv 0$, confirm $\eta^\star$ arbitrary and R_ρ reduces to the log-normalizer term. (iii) Under independence (negative control), $\eta^\star \approx 0$ and the Hopf residual dominates the objective.

Minimal config stub (to mirror your Section 10 API).

```
config := \{
  dX: ground\_metric\_on\_X,\quad
  P\_theta: ef\_model,\quad
  lambda: stratified\_reference\_measure,\quad
  strata\_policy: same\_stratum\_by\_default,\quad
  c\_radius: 1.0,\quad
  t\_id: window\_index,\quad
  n\_eff: effective\_sample\_size,\quad
  d\_eff: None\ |\ estimated\_transport\_dimension,\quad
  M\_mc: 1024,\quad
  batch: 2048,\quad
  lambda\_hopf: 1e-2
\}
```

5 Entropy and Mixed Information Flow

We fix the EF neighborhood and prove a *coordinate-invariant* local equivalence between D_{KL} and the Fisher-metric norm of mixed cumulants. New in this draft: (i) an explicit radius via a Hessian-Lipschitz constant and sharp constants (Theorem 2); (ii) a conditioning corollary (Corollary 2); (iii) a windowed/transported whitening policy with bounds (Proposition 7).

5.1 Dictionary and EF model

Let $\mathcal{T}_K$ be a finite, centered, whitened dictionary of alternated mixed statistics up to total degree $K \in \{2, 3\}$ at *independence* $P_0 = P_\Gamma \otimes P_G$. Denote $F := \mathrm{FIM}(0) \succ 0$ with respect to $\mathcal{T}_K$. We use the EF

$$\frac{dP_\theta}{dP_0}(x) = \exp\big(\langle \theta, \mathcal{T}_K(x)\rangle - \psi(\theta)\big), \qquad \kappa_{\mathrm{mix}}(\theta) = \mathbb{E}_{P_\theta}[\mathcal{T}_K] = \nabla\psi(\theta).$$

Theorem 2 (FIM-invariant local equivalence)**.** [Inf] *Suppose ψ is twice differentiable on a neighborhood of 0 with $\nabla^2\psi(0) = F \succ 0$ and Hessian-Lipschitz with constant L_ψ there. For any $0 < \epsilon < 1$ and radius*

$$r \;\le\; \frac{\epsilon}{L_\psi},$$

we have, for all $\|\theta\| \le r$,

$$\frac{1}{2}(1-\epsilon)\,\|\kappa_{\mathrm{mix}}(\theta)\|^2_{F^{-1}} \;\le\; D_{\mathrm{KL}}(P_\theta \| P_0) \;\le\; \frac{1}{2}(1+\epsilon)\,\|\kappa_{\mathrm{mix}}(\theta)\|^2_{F^{-1}}.$$

Uses: (C5) $F \succ 0$, L_ψ, tolerance ϵ, radius $r \le \epsilon/L_\psi$.

Lemma 10 (Third-moment bound for L_ψ)**.** *For a whitened dictionary $\mathcal{T}_K$ and $F = \nabla^2\psi(0)$, there exist absolute $C > 0$ such that, for $\|u\| = 1$ and small $\|\theta\|$,*

$$\big\|\nabla^2\psi(\theta) - F\big\| \;\le\; C \sup_{\|u\|=1} \mathbb{E}_{P_0}\Big[\big|\langle u, T_K\rangle\big|^3\Big] \cdot \|\theta\| \;+\; O(\|\theta\|^2).$$

Hence one may take $L_\psi \lesssim \sup_{\|u\|=1} \mathbb{E}_{P_0}[|\langle u, T_K\rangle|^3]$.

Remark 3 (Estimator). Estimate the proxy via Hutch++/finite-difference probes (your `estimate_Lpsi` API) and log $(L_\psi,\ r = \epsilon/L_\psi)$ with a CI.

Remark 4 (Tight constants and how to pick them). The $1/2(1 \pm \epsilon)$ constants are optimal for quadratic ψ and remain sharp to first order in ϵ by Taylor's theorem with integral remainder and $\|\nabla^2\psi(\theta) - F\| \le L_\psi\|\theta\| \le \epsilon$.

Corollary 2 (Conditioning in unwhitened coordinates)**.** [Inf] *Let $\tilde\theta = R\theta$ for an invertible linear map with Fisher congruence $\tilde F = R^{-\top} F R^{-1}$. If $\chi = \kappa(F)$ is the condition number of F, then for the same ϵ and r,*

$$\frac{1}{2\chi}(1-\epsilon)\,\|\kappa_{\mathrm{mix}}(\tilde\theta)\|^2_{\tilde F^{-1}} \;\le\; D_{\mathrm{KL}}(P_{\tilde\theta}\|P_0) \;\le\; \frac{\chi}{2}(1+\epsilon)\,\|\kappa_{\mathrm{mix}}(\tilde\theta)\|^2_{\tilde F^{-1}}.$$

Uses: (C5) $\chi = \kappa(F)$; Fisher congruence under coordinate changes.

5.2 Windowed FIM and whitening across time

Let $\mu_{[t-\tau,t]}$ be the windowed law from Section 3. Denote by $F_t := \mathrm{FIM}_t(0)$ the Fisher matrix of the *same* dictionary $\mathcal{T}_K$ computed at independence within that window. Two whitening policies are admissible; pick one and use it consistently.

Definition 6 (Whitening policies)**.** *[Inf/Stoch]*

1. **Global whitening (baseline).** Fix $F := \mathrm{FIM}_{t_0}(0)$ at a reference window (or at the pooled baseline) and keep it fixed for all windows.
2. **Transported whitening (congruence).** For each window, whiten by F_t and transport coordinates to the baseline via the congruence map R_t defined by $F_t = R_t^{-\top} F R_t^{-1}$ (e.g., $R_t = F^{1/2} U_t \Lambda_t^{-1/2} U_t^\top$ with $F_t = U_t \Lambda_t U_t^\top$).

Define the *whitening leakage* $\chi(t) := \kappa(F_t F^{-1})$ and $\chi_{\max} := \sup_t \chi(t)$ on the time span of interest. *Budget & alarms:* enforce $\chi(t) \le \chi_{\max}$ (configuration) and log any exceedance; report $\chi_{\max}$ next to CI bars for time-series plots.

Uses: (C4) window length τ; (C5) baseline F, windowwise F_t, $\chi(t)$.

Proposition 7 (Window-local equivalence and transported bounds)**.** [Inf/Stoch] *Assume each window admits an EF neighborhood with the* same *Hessian-Lipschitz constant L_ψ for ψ_t on a fixed ball and that $F_t \succ 0$. Then, for either whitening policy in Definition 6:*

1. *(*Window-local equivalence*) For all $\|\theta\| \le r \le \epsilon/L_\psi$,*

$$\frac{1}{2}(1-\epsilon)\,\|\kappa^{(t)}_{\mathrm{mix}}(\theta)\|^2_{F_t^{-1}} \;\le\; D_{\mathrm{KL}}(P^{(t)}_\theta\|P^{(t)}_0) \;\le\; \frac{1}{2}(1+\epsilon)\,\|\kappa^{(t)}_{\mathrm{mix}}(\theta)\|^2_{F_t^{-1}}.$$

2. *(*Transport to baseline*) Expressing window-t cumulants in the baseline Fisher norm incurs at most $\chi(t)$ inflation:*

$$\frac{1}{2\,\chi(t)}(1-\epsilon)\,\|\kappa^{(t)}_{\mathrm{mix}}(\theta)\|^2_{F^{-1}} \;\le\; D_{\mathrm{KL}}(P^{(t)}_\theta\|P^{(t)}_0) \;\le\; \frac{\chi(t)}{2}(1+\epsilon)\,\|\kappa^{(t)}_{\mathrm{mix}}(\theta)\|^2_{F^{-1}}.$$

If $\chi(t) \le \chi_{\max} < \infty$ uniformly (e.g. under slow drift and bounded transport dimension), then the constants are uniform in t with χ replaced by $\chi_{\max}$.

Uses: (C4) drift/mixing uniformity; (C5) F_t, L_ψ, $\chi(t)$.

Practical policy. Prefer *global whitening* when interpretability across time is crucial (fixed norm); prefer *transported whitening* when per-window conditioning is poor (reduces numerical ill-conditioning at the cost of tracking $\chi(t)$).

5.3 Diagnostics and parameter selection

Rule (Choosing ϵ and r). Pick $\epsilon \in [0.05, 0.2]$ to balance tightness vs. radius; set $r = \epsilon / L_\psi$ using a pilot bound on L_ψ from finite differences of $\nabla^2 \psi$ at $\theta = 0$ (or Hutch++ trace bounds). Log (ϵ, r, L_ψ) alongside $\chi(t)$ for each window.

Acceptance checks for Section 5.

1. Every "local KL–cumulant" statement cites Theorem 2 with explicit (ϵ, r) and F.
2. Any computation outside whitened coordinates cites Corollary 2 (or Proposition 7 in windows) and reports the relevant χ.
3. Windowed analyses declare the whitening policy and either bound $\chi_{\max}$ or include it in reported uncertainty.

Numerical API (EF radius, L_ψ, whitening leakage $\chi(t)$)

```
# Assumptions
# - T(x): centered, whitened sufficient-statistic vector (same dictionary across windows)
# - Baseline Fisher F (SPD) is fixed (global whitening) or a chosen reference
# - window data: {x_i}_{i=1}^n with precomputed T_i := T(x_i)

LinearAlgebra:
  chol(SPD M) -> L s.t. M = L L^T
  solve_SPD(M, b) -> x with M x = b using Cholesky
  power_norm(A_mv, dim, iters=10) -> approx spectral norm via power iteration on mv-product
  hutchpp_opnorm(A_mv, dim, s=4, iters=2) -> Hutch++-style operator-norm proxy (optional)

# ---------- Window Fisher and whitening leakage ----------
function fisher_window(T_list):
    # Empirical Fisher at independence in this window (covariance of T)
    # ridge for numerical stability (lambda_ridge small, e.g. 1e-6 * trace/dim)
    n = len(T_list); d = dim(T_list[0])
    mu = mean(T_list)                          # should be ~0 by centering; keep for safety
    C  = (1/n) * sum_i (T_list[i] - mu) (T_list[i] - mu)^T
    lam = 1e-6 * trace(C) / d
    return C + lam * I_d

function chi_leakage(F_baseline, F_t):
    # Compute chi(t) = kappa(F_t F_baseline^{-1}) via generalized EVP
    # Equivalent to eigenvalues of S := F_baseline^{-1/2} F_t F_baseline^{-1/2}
    L = chol(F_baseline)                       # F = L L^T
    # mv for S: v -> L^{-1}^T * F_t * L^{-1} * v  (apply solves with L and L^T)
    function S_mv(v):
        y = solve_SPD(L, v)                    # L y = v
        z = F_t @ y
        w = solve_SPD(L.T, z)                  # L^T w = z
        return w
    # estimate extremal eigenvalues of S by power and inverse power
    # power_norm returns ||S||_2^{1/2} if fed S_mv? -> square it
    lam_max = power_norm(S_mv, dim=d, iters=12)^2
    # For lam_min, run power on inverse of S via CG solves (or Gershgorin fallback):
    # implement inverse-power via Rayleigh quotient iteration on (S + tau I)^{-1} if needed.
    # or use eigs on small d
    lam_min = 1.0 / power_norm(lambda v: cg_solve(S_mv, v), dim=d, iters=12)^2
    return lam_max / lam_min

```

```
# ---------- Small-tilt Hessian probes and L_psi estimation ----------
# EF identity: \nabla^2 psi(\theta) = Cov_{P_theta}(T). Around theta=0, estimate Lipschitz via:
#   || \nabla^2 psi(\delta u) - \nabla^2 psi(0) || / ||\delta u||.
# We approximate P_{delta u} by reweighting samples from P_0 with weights w_i \prop exp(\delta <u, T_i>).

function weighted_cov(T_list, u, delta):
    s = [ dot(u, T_i) for T_i in T_list ]
    a = max(s)                          # log-sum-exp stabilization
    w_unn = [ exp(delta * (s_i - a)) for s_i in s ]
    Z  = sum(w_unn); w = [ w_unn_i / Z for w_unn_i in w_unn ]
    mu_w = sum_i w[i] * T_list[i]
    C_w  = sum_i w[i] * (T_list[i] - mu_w) (T_list[i] - mu_w)^T
    return C_w

function mv_diff_H(T_list, H0, u, delta):
    # returns an mv-product for A := (H(delta u) - H0)/delta without forming it
    H_delta = weighted_cov(T_list, u, delta)
    A = (H_delta - H0) / delta
    return (lambda v: A @ v)

function estimate_Lpsi(T_list, deltas=[1e-3, 3e-3, 1e-2], m_dirs=8, use_hutchpp=True):
    d  = dim(T_list[0])
    H0 = fisher_window(T_list)                   # \nabla^2 psi(0) at independence
    Lpsi_hat = 0.0
    for k in 1..m_dirs:
        u0 = gaussian(d); u = u0 / norm(u0)     # random unit direction (Euclidean or F-based)
        for delta in deltas:
            A_mv = mv_diff_H(T_list, H0, u, delta)  # mv for (H(delta u) - H0)/delta
            if use_hutchpp:
                opn = hutchpp_opnorm(A_mv, dim=d, s=4, iters=2)
            else:
                opn = power_norm(A_mv, dim=d, iters=10)
            Lpsi_hat = max(Lpsi_hat, opn)
    return Lpsi_hat, H0

# ---------- Radius selection and logging ----------
function pick_radius(epsilon, Lpsi_hat, floor=1e-8):
    Lp = max(Lpsi_hat, floor)
    r  = epsilon / Lp
    return r

function section5_online_update(window_T, F_baseline, epsilon, log_json):
    # 1) window Fisher and leakage
    F_t  = fisher_window(window_T)
    chi  = chi_leakage(F_baseline, F_t)
    # 2) Lipschitz estimate and EF radius
    (Lpsi_hat, H0) = estimate_Lpsi(window_T)
    r = pick_radius(epsilon, Lpsi_hat)
    # 3) structured logging (aligns with section 5 acceptance checks)
    log_json({
        "section": "5",
        "epsilon": epsilon,
        "Lpsi_hat": Lpsi_hat,
        "r_radius": r,
        "chi_t": chi,
        "trace_F_t": trace(F_t),
        "cond_F_t": cond(F_t),              # optional: standard cond estimate
        "n_stats": len(window_T),
        "d_dict": dim(window_T[0])
```

```
    })
    return {"F_t": F_t, "chi": chi, "Lpsi_hat": Lpsi_hat, "r": r, "H0": H0}

# ---------- Guards and unit checks ----------
# - If chi explodes (e.g., > 1e6), switch to transported whitening (section 5.2) and re-estimate.
# - If r is too small (below step size of optimizer), cap epsilon upward or enrich dictionary to reduce Lpsi_hat.
# - Validate: for synthetic quadratic psi, Lpsi_hat -> 0 and r -> infty (subject to floor).
```

Notes and good practice.

- *Direction metric.* Draw u as unit vectors in the *Fisher* norm if you want L_ψ in the induced operator norm of F^{-1}; otherwise Euclidean is fine—just be consistent with the norm used in Theorem 2.
- *Small-tilt accuracy.* The reweighting step is accurate for small δ; keep $\delta \in [10^{-3}, 10^{-2}]$ (rescale if $\|T\|$ is large) and monitor ESS $= 1/\sum w_i^2$.
- *Stability.* Always ridge F_t lightly before Cholesky; report the ridge size in logs if it exceeds $10^{-6} \times \mathrm{trace}(F_t)/d$.
- *Throughput.* Cache T_i once per window; reuse them for both F_t and L_ψ estimation. Hutch++ gives a robust operator-norm proxy with a handful of matvecs.

6 Entropy-Aware Projected Control

We formalize a Dirac projection Π and band-limited fractional controllers. New here: (i) a proof that D_i is a Dirac structure (maximally isotropic) under rank-regularity (Lemma 7); (ii) a *frontier freeze* rule; (iii) verifiable PR conditions via KYP or phase-margin bounds (Lemmas **??**–10); (iv) an ISS theorem with an *explicit* frontier term scaling as $t/\tau_{\min}$ (Theorem 3); (v) entropy-aware gain adaptation via either an admissible-set projection (KYP) or a two-time-scale separation.

6.1 Port-Hamiltonian plant and Dirac projection

On each stratum M_i, a port-Hamiltonian form reads

$$\dot{x} \in J_i(x)\sharp dH(x) - R_i(x)\sharp dH(x) + G_i(x)u, \qquad y = G_i(x)^* dH(x),$$

with skew $J_i = -J_i^*$, positive semidefinite $R_i \succeq 0$, and storage $H \geq 0$.

Definition 6 (Dirac projection from strategic fields)**.** Given $\hat{S}$ and a linear operator $\mathbb{A}_i(x; G) : T_x\Gamma \to \mathbb{R}^r$ (from the J-aligned plane selection of Section 7), define

$$\mathcal{X}_i(x) := \ker \mathbb{A}_i(x; G), \qquad \mathcal{X}_i^0(x) := \{e \in T_x^*\Gamma : \ e(f) = 0 \ \forall f \in \mathcal{X}_i(x)\}.$$

Set the Dirac structure $D_i(x) := \mathcal{X}_i(x) \oplus \mathcal{X}_i^0(x) \subset T_x\Gamma \oplus T_x^*\Gamma$ and let $\Pi_i(x)$ be the orthogonal projector onto $D_i(x)$ for the energetic pairing $\langle e, f\rangle$.

Lemma 7 (Maximal isotropy and smoothness of D_i)**.** [Ctrl/Rep] *If $\mathbb{A}_i(\cdot; G)$ has constant rank on M_i (rank drops only on frontiers), then $D_i := \mathcal{X}_i \oplus \mathcal{X}_i^0$ is maximally isotropic in $T\Gamma \oplus T^*\Gamma$ and forms a smooth subbundle on M_i. Consequently, Π_i depends smoothly on x along strata.*

Uses: (C1) rank-regularity per stratum; annihilator geometry.

Lemma 8 (Supply-rate preservation)**.** [Ctrl] *Orthogonal projection onto a maximally isotropic subspace preserves the supplied power: if $\Pi_i((f, e), (u, y)) = ((\tilde{f}, \tilde{e}), (\tilde{u}, \tilde{y}))$ then*

$$\langle e, f\rangle + u^\top y \ = \ \langle \tilde{e}, \tilde{f}\rangle + \tilde{u}^\top \tilde{y}.$$

Uses: Energetic pairing; Dirac isotropy.

Rule (Frontier projection freeze). During an instantaneous stratum crossing (frontier tube), hold Π fixed at its pre-crossing value; reproject immediately after the crossing.

Uses: (C3) dwell $\tau_{\min}$ from Lemma 2; (C1) frontier tube; (C2) bounded fields.

6.2 Band-limited fractional controllers and PR checks

Let $C_N(s)$ be an Nth-order Oustaloup approximation of s^α on $[\omega_L, \omega_H]$ with a state-space realization (A, B, C, D) after bilinear (band) mapping.

Lemma 9 (GKYP sufficiency for band-limited PR)**.** [Ctrl] *Let (A, B, C, D) realize $C_N(s)$ after the bilinear band map. There exist $P = P^\top \succ 0$ and a Hermitian multiplier $Q = Q^\top \succeq 0$ such that the frequency-interval GKYP LMI*

$$\begin{bmatrix} A & B \\ I & 0 \end{bmatrix}^\top (\Xi(\omega_L, \omega_H) \otimes P) \begin{bmatrix} A & B \\ I & 0 \end{bmatrix} \;+\; \begin{bmatrix} C & D \end{bmatrix}^\top \Pi_{\mathrm{PR}} \begin{bmatrix} C & D \end{bmatrix} \;\preceq\; -\epsilon I$$

holds, where $\Xi(\omega_L, \omega_H)$ is the standard frequency-interval multiplier and $\Pi_{\mathrm{PR}} = \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}$ encodes positive realness. Then C_N is $(\epsilon, [\omega_L, \omega_H])$-positive real. Note: *the bilinear band map recovers the same inequality on the unit circle.*

Uses: (C6) PR margin ϵ; GKYP (frequency-interval) multipliers.

Lemma 10 (Phase-margin sufficiency)**.** [Ctrl] *If $\max_{\omega \in [\omega_L, \omega_H]} |\arg C_N(j\omega) - \alpha\pi/2| < \frac{\pi}{2} - \delta$ and* DC *gain $K_0 > 0$, then $\Re\, C_N(j\omega) \geq \epsilon(\delta, K_0) > 0$ on the band.*

Uses: (C6) phase-error bound $\Rightarrow$ PR margin $\epsilon(\delta, K_0)$.

Admissible gain policy (verifiable). Let $g \in \mathbb{R}^p$ denote controller gains (e.g., scalings of C_N sections). Define the *PR-admissible set*

$$\mathcal{G} \;:=\; \{\, g :\; \exists\, P \succ 0 \text{ s.t. } \mathrm{KYP}(A(g), B(g), C(g), D(g); P) \text{ holds on } [\omega_L, \omega_H] \,\}.$$

At run time, set $g \leftarrow \mathrm{proj}_{\mathcal{G}}(g_{\mathrm{nom}} \cdot \phi(\|\kappa_{\mathrm{mix}}\|_{\mathrm{FIM}^{-1}}))$ with a Lipschitz, monotone $\phi(\cdot)$; log the achieved PR margin from Lemma **??** for audit.

Two-time-scale alternative. Assume *(TTS)*: $\dot{g} = \varepsilon\, \Xi(\|\kappa_{\mathrm{mix}}\|)$ with $0 < \varepsilon \ll 1$ (slower than plant), and $g(t)$ remains within an open PR band for all t. Then the PR margin is preserved by averaging.

6.3 Closed-loop interconnection and ISS under switching

Lemma 11 (Storage jump at a crossing)**.** *Across any instantaneous crossing t_k, the storage satisfies*

$$H(x(t_k^+)) - H(x(t_k^-)) \;\leq\; C_{\mathrm{fr}},$$

with C_{fr} from Lemma 4. Summing the jumps and using Lemma 2 yields the $t/\tau_{\min}$ contribution in the ISS bound.

Theorem 3 (Passivity $\Rightarrow$ ISS with frontier defects)**.** [Ctrl] *Suppose on each stratum M_i: (i) there exists a storage H with $\dot{H} \leq y^\top u - \alpha\|y\|^2$; (ii) interconnection via Π preserves power (Lemma 8); (iii) C_N is $(\epsilon, [\omega_L, \omega_H])$-PR (Lemma **??** or 10); (iv) frontier crossings are finite on finite horizons (Lemma 2); (v) the frontier defect obeys Lemma 4. Then the closed loop is ISS and, for all $t \geq 0$,*

$$\boxed{H(x(t)) \;\leq\; \beta\big(H(x(0))\big) \;+\; \int_0^t \gamma\big(\|w(\tau)\|\big)\, d\tau \;+\; C_0\, N_{\mathrm{cross}}(0, t)\,,}$$

with $C_0 = C_0\big(S_{\max}, G_{\max}, \kappa_{\max}, \theta_{\min}\big)$ *as in Lem. 4. By Lem. 2,* $N_{\mathrm{cross}}(0,t) \le t/\tau_{\min}$*, hence*

$$\boxed{H(x(t)) \;\le\; \beta(H(x(0))) + \int_0^t \gamma(\|w(\tau)\|)\, d\tau \;+\; \frac{C_0}{\tau_{\min}}\, t.}$$

Uses: $\tau_{\min}$ *(No-Zeno), geometry-scaled defect* $C_0(\cdot)$*, PR margin* ϵ*.*

Entropy-aware gains (guaranteed variants).

Projection policy (preferred).
Use the admissible set $\mathcal{G}$; the gain update is $g \leftarrow \mathrm{proj}_{\mathcal{G}}(g_{\mathrm{nom}} \cdot \phi(\|\kappa_{\mathrm{mix}}\|_{\mathrm{FIM}^{-1}}))$. Since $\mathcal{G}$ is PR-certified by KYP, the PR margin ϵ is preserved pointwise in time, and Theorem 3 applies verbatim.

Two-time-scale policy.
Under *(TTS)*, average the dissipation inequality over windows of length $\gg 1/\varepsilon$; the effective margin is $\epsilon + O(\varepsilon)$, hence ISS with slightly perturbed constants.

Design/implementation checklist (auto-audit).

- *Dirac* II: log rank of $\mathbb{A}_i$, detect/flag frontier freeze events, and record crossing times.
- *PR checks*: implement `check_positive_real(C_N, band, ϵ)` using Lemma **??** (primary) or Lemma 10 (fallback); log the achieved margin per window.
- *Entropy-aware*: record $\|\kappa_{\mathrm{mix}}\|_{\mathrm{FIM}^{-1}}$, chosen g, whether projection to $\mathcal{G}$ occurred, and the *(TTS)* flag if used.
- *ISS budget*: report α, ϵ, C_0, $N_{\mathrm{cross}}(0,t)$ and its bound $t/\tau_{\min}$ alongside $H(x(t))$.

Acceptance checks for Section 6.

1. The ISS inequality shows the frontier term as $C_0 N_{\mathrm{cross}}(0,t)$ and then substitutes $N_{\mathrm{cross}} \le t/\tau_{\min}$.
2. Any use of "entropy-aware gains" references either the admissible-set projection (`proj_`$\mathcal{G}$) or the two-time-scale assumption; PR preservation is explicit.
3. Logs include PR margin on $[\omega_L, \omega_H]$, counts of frontier freezes/crossings, and the realized $t/\tau_{\min}$ bound.

Numerical API (Dirac, PR checks, gain projection, ISS budget)

```
# --- Inputs (conceptual types) ---------------------------------------------
Dirac:
  A_i(x;G)                      # per-stratum constraint operator (Definition 6)
  build_dirac_projection(A_i)  # returns Pi_i(x) with energetic inner product

Controller:
  oustaloup_ss(alpha, N, band, K) -> (A,B,C,D)   # state-space for C_N(s)=K s^alpha approx on [wL,wH]
  band = [wL, wH]

Metrics:
  energy_pairing(e,f) = <e,f>   # energetic pairing
  H(x)                          # storage
  kappa_mix_norm := ||kappa_mix||_{FIM^{-1}}     # from Section~5/Section~3

```

```
Logging:
  log_json(dict)
  now() -> t

# --- Frontier handling ------------------------------------------------------
state frontier = { crossing:false, last_cross_t:None, N_cross:0 }

function on_frontier_enter():
    frontier.crossing = true
    frontier.N_cross += 1
    frontier.last_cross_t = now()
    freeze_Pi_and_controller_state()         # Rule: Frontier projection freeze (Section 6)
    log_json({"evt":"frontier_enter", "t": frontier.last_cross_t, "N_cross": frontier.N_cross})

function on_frontier_exit():
    frontier.crossing = false
    reproject_Pi_post_event()                # recompute Pi on new stratum
    log_json({"evt":"frontier_exit", "t": now(), "N_cross": frontier.N_cross})

# --- KYP positive-real checker (band-limited) -------------------------------
# Maps [wL,wH] to the unit circle via a bilinear band map, then runs a KYP LMI.
# If an SDP solver is unavailable, falls back to dense frequency sampling + phase-margin test.

function check_PR_band(A,B,C,D, band, eps_min=0.0, sdp_solver=None, grid_pts=512):
    (Aw,Bw,Cw,Dw) = bilinear_band_map(A,B,C,D, band)  # stable map of band to unit circle
    if sdp_solver is not None:
        # Continuous-time PR LMI (after mapping the band):
        # Find P >> 0 and epsilon >= 0 s.t.
        #   [A^T P + P A,      P B - C^T;
        #    B^T P - C,    -(D + D^T)]  <=  -epsilon I
        (feasible, P, eps) = sdp_solver_PR(Aw,Bw,Cw,Dw)
        if feasible and eps >= eps_min:
            return {"pr_ok": True, "eps": eps, "cert":"KYP", "P": P}
    # Fallback: phase-margin sufficiency (Lemma 10)
    w = logspace(log10(band[0]), log10(band[1]), grid_pts)
    phi_err_max = 0.0; min_Re = +inf
    for wi in w:
        H = freqresp_ct(A,B,C,D, wi)
        min_Re = min(min_Re, Re(H))
        phi_err_max = max(phi_err_max, abs(arg(H) - alpha*pi/2))
    if min_Re > 0:
        eps = min_Re                              # crude PR margin proxy
        return {"pr_ok": True, "eps": eps, "cert":"phase", "phi_err_max": phi_err_max}
    return {"pr_ok": False, "eps": 0.0, "cert":"none", "phi_err_max": phi_err_max}

# --- Admissible gain set projection  G  -------------------------------------
# If C_N depends affinely on gains g (common in cascaded Oustaloup sections), the set
#   G = { g : exists P >> 0 s.t. KYP(A(g),B(g),C(g),D(g);P) holds on [wL,wH] }
# is convex. We implement a practical projection by:
#   (1) try KYP feasibility at g; if infeasible, (2) bisection on a scalar scale s in [s_min,s_max]
#       toward a feasible point; (3) optional convex QP if an affine parametrization is available.

function project_to_G(g_nom, phi_kappa, model, band, sdp_solver, s_bounds=[0.3, 2.0], tol=1e-3):
    g_target = g_nom * phi_kappa              # entropy-aware scaling
    if KYP_feasible(model, g_target, band, sdp_solver):
        return g_target, {"projected": False, "scale": 1.0}
    # Bisection on scalar scale s to nearest feasible:
    s_lo, s_hi = s_bounds[0], s_bounds[1]
    # Ensure bracket: expand outward if necessary
```

```
    for it in 1..6:
        if KYP_feasible(model, g_nom*s_lo, band, sdp_solver): break
        s_lo *= 0.7
    for it in 1..6:
        if KYP_feasible(model, g_nom*s_hi, band, sdp_solver): break
        s_hi *= 1.4
    if not KYP_feasible(model, g_nom*s_hi, band, sdp_solver):
        # give up: return last feasible baseline (conservative)
        return g_nom, {"projected": True, "scale": 1.0, "note":"fallback_to_baseline"}
    # Bisection to tolerance
    for it in 1..20:
        s_mid = 0.5*(s_lo + s_hi)
        if KYP_feasible(model, g_nom*s_mid, band, sdp_solver):
            s_hi = s_mid
        else:
            s_lo = s_mid
        if abs(s_hi - s_lo) < tol * s_hi: break
    g_feas = g_nom * s_hi
    return g_feas, {"projected": True, "scale": s_hi}

# Helper: feasibility wrapper
function KYP_feasible(model, g, band, sdp_solver):
    (A,B,C,D) = model.ss(g)                    # affine in g
    res = check_PR_band(A,B,C,D, band, eps_min=0.0, sdp_solver=sdp_solver)
    return res["pr_ok"]

# --- Runtime controller synthesis & audit -------------------------------------
function controller_step(alpha, N, band, g_nom, kappa_mix_norm, model, sdp_solver):
    phi = phi_entropy(kappa_mix_norm)          # monotone Lipschitz policy
    (g, proj_info) = project_to_G(g_nom, phi, model, band, sdp_solver)
    (A,B,C,D) = model.ss(g)
    pr = check_PR_band(A,B,C,D, band, eps_min=0.0, sdp_solver=sdp_solver)
    log_json({"evt":"ctrl_update",
              "kappa_mix_norm": kappa_mix_norm,
              "phi": phi,
              "g_nom": g_nom, "g_used": g,
              "proj": proj_info, "pr_cert": pr["cert"], "pr_eps": pr["eps"]})
    return (A,B,C,D), pr

# --- ISS budget tracker (explicit frontier term) ------------------------------
state iss = { H0: None, alpha_bar: None, eps_PR: None, C0: None, tau_min: None }

function iss_init(H0, alpha_bar, eps_PR, C0, tau_min):
    iss.H0 = H0; iss.alpha_bar = alpha_bar; iss.eps_PR = eps_PR; iss.C0 = C0; iss.tau_min = tau_min

function iss_bound(t, input_energy_int):
    # H(x(t)) <= e^{-alpha_bar t} H0 + input_energy_int + C0 * N_cross(0,t)
    N = frontier.N_cross
    bound_switch = iss.C0 * N
    bound_linear = (iss.C0 / iss.tau_min) * t
    H_bound = exp(-iss.alpha_bar * t) * iss.H0 + input_energy_int + bound_switch
    log_json({"evt":"iss_budget",
              "t": t, "H_bound": H_bound,
              "N_cross": N, "C0": iss.C0,
              "tau_min": iss.tau_min, "frontier_term": bound_switch,
              "frontier_term_linear": bound_linear})
    return H_bound, {"frontier_term": bound_switch, "frontier_term_linear": bound_linear}

# --- One-call loop integration (per control tick) -----------------------------
```

```
function control_tick(x, env, model, params):
    if frontier_event_detected(env): on_frontier_enter()
    (A,B,C,D), pr = controller_step(params.alpha, params.N, params.band,
                                    params.g_nom, kappa_mix_norm(x),
                                    model, params.sdp_solver)
    if frontier.crossing:             # freeze rule during instantaneous crossing
        hold_controller_state()       # keep last (A,B,C,D) until exit
    u = apply_dirac_and_controller(x, A,B,C,D)   # \Pi from build_dirac_projection(A_i)
    env.step(u)
    if frontier_exit_detected(env): on_frontier_exit()
    # ISS audit
    H_t = H(x)
    input_energy = params.input_integral.update(y_times_u=...)    # accumulate \int_0^t \gamma(||w||) d\tau if avai
    H_bound, ft = iss_bound(t=now(), input_energy_int=input_energy)
    # Periodic audit log
    log_json({"evt":"tick", "t": now(), "H": H_t, "H_bound": H_bound,
              "pr_eps": pr["eps"], "N_cross": frontier.N_cross, "k_plane": params.k_plane_mode})
```

Guards, tests, and good practice. - **Bilinear map sanity.** Validate the band map with a unit-gain integrator: the mapped discrete pole should land exactly on the unit circle at the designated band center. - **SDP fallback.** If an SDP solver isn't available, the phase-margin grid (Lemma 10) is a *sufficient* test. Log 'pr_cert â^^ KYP, phase, none' so failures are visible. - **Projection stability.** If repeated projections keep shrinking gains, widen the band or increase N (Oustaloup order) to reduce phase error; both actions typically expand $\mathcal{G}$. - **Freeze discipline.** During any flagged crossing, hold Π and controller internal state (integrators/filters) fixed, then reproject immediately after exit—this is your boxed rule. - **Audit completeness.** Each window should have: 'pr_eps', 'N_cross', 'tau_min', 'C0', and the bound terms. These tie directly to Theorem 3 and your acceptance checks.

Mini unit tests. 1) **PR checker:** For a known PR network (e.g., $C(s) = K/(1+s)$) on a narrow band, KYP must return 'pr_ok=True' and $\epsilon > 0$. 2) **Projection:** Intentionally scale gains to violate PR; 'project_to_G' should return a scaled g with 'projected=True' and 'check_PR_band(...)=True'. 3) **ISS budget:** Simulate a run with zero input and no crossings; ensure the logged bound decays as $e^{-\underline{\alpha}t}$ and the frontier terms remain zero; then inject synthetic crossings and verify the linear-in-t bound via $\tau_{\min}$.

"Uses:" (implicit for Section 6). *Uses:* (C6) PR margin; (C3) $\tau_{\min}$ from Lemma 1; (C7) C_{fr} via Lemma 3; energetic pairing; band $[\omega_L, \omega_H]$; Oustaloup order N; freeze rule.

7 Computable Kernels: Plane Selection and Structural Derivatives

We turn abstract circulation bivectors into numerically stable kernels for control/estimation. New here: (i) admissible constrained subbundles for plane selection with upper semicontinuity of maximizers and a k-plane degeneracy policy; (ii) SVD selection aligned with the *almost-complex* structure of real skew matrices; (iii) explicit regularity and finite-difference (FD) policies for shape derivatives with a *frontier freeze* and dependence-aware CIs.

7.1 Plane selection under constraints: skew geometry and robustness

Let $A(\gamma, G)$ be the real skew matrix representing the Γ-circulation $\Phi = S_\Gamma^{(2,0)}$. Let $\mathcal{U}(\gamma, G) \subset \mathbb{R}^d$ and $\mathcal{V}(\gamma, G) \subset \mathbb{R}^d$ be *admissible subspaces* encoding feasibility (e.g., contact modes, actuator limits). We assume:

Assumption 2 (Admissible subbundles and frontier rule)**.** *[Rep/Ctrl]* (i) On each stratum M_i, $\mathcal{U}$ and $\mathcal{V}$ are C^1 subbundles with orthogonal projectors $P_U(\gamma, G)$ and $P_V(\gamma, G)$ depending continuously on (γ, G); (ii) across a frontier crossing, apply the *freeze* rule of Section 6: hold P_U, P_V fixed during the crossing and re-evaluate immediately after.

Uses: (C1) stratified smoothness; frontier freeze (Rule in Section 6).

For a given (γ, G), define the constrained operator $A_{UV} := P_U A P_V$.

Lemma 11 (Constrained SVD)**.** [Alg/Ctrl] *The maximizers of*

$$\max_{\substack{X\in\mathcal{U},\ \|X\|=1\\ Y\in\mathcal{V},\ \|Y\|=1}} |X^\top A Y|$$

are the top left/right singular vectors of A_{UV}; the optimum equals $\sigma_1(A_{UV})$.

Uses: Orthogonal projectors P_U, P_V; induced operator norm.

Lemma 12 (Upper semicontinuity of maximizing subspaces)**.** [Rep/Alg] *If $(\gamma_k, G_k) \to (\gamma, G)$ within a stratum and Assumption 2 holds, then the set of maximizers of Lemma 11 is* upper semicontinuous*: any limit of maximizers at (γ_k, G_k) is a maximizer at (γ, G). Consequently, with the frontier freeze, the selected subspaces are piecewise continuous and free of chattering.*

Proof sketch. Continuity of P_U, P_V gives continuity of A_{UV}; the singular-value map is continuous and the argmax correspondence is upper semicontinuous on compact Grassmannians. The freeze eliminates instantaneous toggling at frontiers. □

Uses: (C1) continuity of P_U, P_V; compactness of Grassmannians; frontier freeze.

Proposition 2 (Freeze-bias for structural derivatives)**.** *Let the total freeze time inside tubes on a window be T_{frz} and the window length be τ. Then for the windowed gradient,*

$$\mathrm{Bias}\left(\widehat{\partial_\theta \mathcal{L}}\right) \ \le\ C\,\frac{T_{\mathrm{frz}}}{\tau} \ +\ O(h^2),$$

with C depending on local drift/Lipschitz moduli. Report: *log T_{frz}/τ with the CI from the dependence-aware bootstrap.*

Remark 5 (Skew-SVD $\Leftrightarrow$ complex eigenplanes). *[*Alg/Ctrl*]* For real skew A, singular values come in pairs and A induces an almost-complex structure J on each principal 2-plane: if $Av = i\lambda v$ with $\lambda > 0$, then the maximizing plane is spanned by $\Re v, \Im v$ and J acts as a $\frac{\pi}{2}$ rotation on it. Under constraints, apply the same construction to A_{UV} on the selected subspace. This J-aligned basis feeds $\mathbb{A}_i$ in Definition 6.

Uses: Skew symmetry; spectral pairing; J-structure.

Degeneracy policy (measurable, robust). Let $\sigma_1 \ge \sigma_2 \ge \cdots$ be the singular values of A_{UV}. If $\sigma_1 - \sigma_2 < \varepsilon_{\mathrm{gap}}$, *escalate* to a k-plane (typically $k = 4$) spanned by the top k singular vectors, and actuate in any J-compatible basis on that plane. With a perturbation Δ and gap $g := \sigma_k - \sigma_{k+1} > 0$, Davis–Kahan/Wedin yields

$$\sin\Theta\big(\widehat{\mathcal{U}}_k, \mathcal{U}_k\big) \ \le\ \frac{\|P_U\,\Delta\,P_V\|}{g},$$

hence the escalation threshold $g \le \varepsilon_{\mathrm{gap}}$ gives a measurable trigger and bounded principal-angle drift under small $\|\Delta\|$.

Uses: Spectral gap g; operator perturbation $\|\Delta\|$; J-compatibility.

7.2 Shape derivatives on G: regularity, FD policy, and freeze

Let $\theta \in \mathbb{R}^p$ parametrize G_θ and $T(\gamma, G_\theta)$ denote a stratumwise functional (e.g., contractions entering A).

Proposition 3 (Score regularity and FD MSE)**.** [Stoch/Inf] *(i) (*Regularity*) If the law admits a stratumwise density $w(\gamma; Y \mid G_\theta)$ differentiable in θ and there exists $g \in L^1$ with $|\nabla_\theta w| \le g$ uniformly on $\|\theta\| \le r$, then*

$$\nabla_\theta \, \mathbb{E}[T(\gamma, G_\theta)] \;=\; \mathbb{E}\big[T \, \nabla_\theta \log w\big] \;+\; \mathbb{E}[\nabla_\theta T],$$

with expectations taken on each stratum and aggregated by λ (Def. 4).
*(ii) (*FD MSE*) Two-sided central differences with step h and m samples per side satisfy*

$$\text{MSE} \;\le\; \frac{2\sigma^2}{mh^2} \;+\; C_b^2 \, h^4, \qquad h^\star \;\asymp\; \Big(\frac{\sigma^2}{m\, C_b^2}\Big)^{1/6}.$$

Proof sketch. (i) Dominated convergence with the L^1 envelope; decompose via the log-derivative trick. (ii) Standard variance $O(m^{-1}h^{-2})$ and truncation bias $O(h^2)$ for central differences give the stated balance. □

Uses: Envelope on $|\nabla_\theta w|$; window laws; variance σ^2; curvature bound C_b.

Rule (Frontier freezing for θ-updates). Hold θ fixed inside frontier tubes and reuse the last valid gradient; re-enable updates after dwell time $\ge \tau_{\min}$ (Lemma 2). The induced bias is $O(\text{dwell})$ and is dominated by window averaging at the same scale.

Uses: (C3) $\tau_{\min}$; (C1) frontier tubes; windowing (Section 3).

Dependence-aware CIs. Report gradient CIs using moving or circular block bootstrap (MBB/CBB) with $b \sim n^{1/3}$ (Section 9). For multivariate gradients, give FIM-weighted ellipsoids; state the chosen b and effective sample size.

Update–reselect kernel with guards (summary). Assemble A, compute A_{UV}, select a J-aligned plane (escalate to k if $\sigma_k - \sigma_{k+1} < \varepsilon_{\text{gap}}$), compute θ-gradients via score or FD with $h^\star$, perform a rank-2 (or $2k$) refresh, and apply the frontier freeze during instantaneous crossings. Export the J-aligned basis to Definition 6 for building $\mathcal{X}_i = \ker \mathbb{A}_i$.

Acceptance checks for Section 7.

1. Plane selection cites Lemma 11 and uses Assumption 2; maximizers are upper semicontinuous (Lemma 12); degeneracy escalation uses the gap trigger and logs principal angles.
2. Skew geometry is explicit (Remark 5); the J-basis exported to Section 6 is stated.
3. Gradient estimators cite Proposition 3; outputs print $h^\star$, bootstrap CI (with block length b), and the freeze status if active.

Numerical API (plane selection, k-plane guard, J-basis, FD $h^\star$, CIs)

```
# === Inputs (conceptual) ==========================================================
Geometry:
  assemble_A(gamma, G, hatS) -> A        # skew matrix for S~{(2,0)}_\Gamma (Sec. 7)
  admissible_subspaces(gamma, G) -> (P_U, P_V)   # orthogonal projectors (Assumption 7.1)

Frontier:
  in_frontier_tube() -> bool             # frontier detector (freezes updates)
  tau_min                                # dwell-time lower bound from Lemma 1

Stats:
  T_stream                               # stream or window of sufficient stats / test functionals
  window_id, n_eff                       # identifiers for logging
```

```
13
14  Logging:
15    log_json(dict)
16
17  # === Helpers: algebra on subspaces ========================================
18  function constrained_svd(A, P_U, P_V, k=2):
19      A_uv = P_U @ A @ P_V
20      # Economy SVD; for skew A_uv, use symmetric structure for speed if available
21      (U, s, Vt) = svd(A_uv, full=False)      # s[0] >= s[1] >= ...
22      return {"U": U[:, :k], "V": Vt.T[:, :k], "s": s[:max(4,k)]}
23
24  function J_aligned_basis(A, U2, V2):
25      # Build a J-compatible 2-plane basis from top singular vectors (skew structure)
26      # For skew A, left/right vectors approximate +/- 90 deg phase shift; orthonormalize
27      QL = qr(U2)[0]; QR = qr(V2)[0]            # 2-col each
28      # Return orthonormal bases for the actuation plane(s)
29      return {"Q_left": QL, "Q_right": QR}
30
31  function principal_angles(Q_prev, Q_new):
32      # Return principal angles between two k-planes (columns orthonormal)
33      # via SVD of Q_prev^T Q_new
34      M = Q_prev.T @ Q_new
35      s = svd(M, compute_uv=False)
36      angles = arccos( clip(s, 0, 1) )
37      return sort_desc(angles)                  # largest first
38
39  # === Degeneracy guard and escalation ======================================
40  state plane_state = { "Q_L": None, "Q_R": None, "sigma": None,
41                        "gap": None, "k": 2, "degenerate": False }
42
43  function plane_select_kernel(gamma, G, hatS, eps_gap=1e-1, k_alt=4):
44      (P_U, P_V) = admissible_subspaces(gamma, G)
45      A = assemble_A(gamma, G, hatS)
46      S2 = constrained_svd(A, P_U, P_V, k=2)
47      s = S2["s"]; gap = s[0] - s[1] if len(s) > 1 else +inf
48      if gap < eps_gap:
49          S4 = constrained_svd(A, P_U, P_V, k=k_alt)
50          B = J_aligned_basis(A, S4["U"], S4["V"])
51          plane_state.k = k_alt; plane_state.degenerate = True
52          plane_state.sigma = S4["s"]; plane_state.gap = S4["s"][k_alt-1] - S4["s"][k_alt] if len(S4["s"])>k_alt else
53          QL, QR = B["Q_left"], B["Q_right"]
54      else:
55          B = J_aligned_basis(A, S2["U"], S2["V"])
56          plane_state.k = 2; plane_state.degenerate = False
57          plane_state.sigma = S2["s"]; plane_state.gap = gap
58          QL, QR = B["Q_left"], B["Q_right"]
59      # principal angles vs previous (for chatter audit)
60      if plane_state["Q_L"] is not None:
61          angL = principal_angles(plane_state["Q_L"], QL)
62          angR = principal_angles(plane_state["Q_R"], QR)
63      else:
64          angL = angR = None
65      plane_state["Q_L"] = QL; plane_state["Q_R"] = QR
66      # log
67      log_json({"evt":"plane_select",
68                "window": window_id,
69                "sigma_top": float(plane_state.sigma[0]),
70                "gap_trigger": float(gap),
71                "k_mode": plane_state.k,
```

```
              "degenerate": plane_state.degenerate,
              "angles_L": angL, "angles_R": angR})
    return {"QL": QL, "QR": QR, "k": plane_state.k, "gap": plane_state.gap,
            "sigma": plane_state.sigma, "angles_L": angL, "angles_R": angR}

# === Frontier freeze hook ================================================
state freeze = { "active": False, "entered_at": None }

function maybe_freeze_updates():
    if in_frontier_tube() and not freeze.active:
        freeze.active = True; freeze.entered_at = now()
        log_json({"evt":"freeze_on", "window": window_id})
    if (not in_frontier_tube()) and freeze.active:
        freeze.active = False
        dwell = now() - freeze.entered_at
        log_json({"evt":"freeze_off", "window": window_id, "dwell": dwell})

# === FD step-size policy (Prop. 2) =======================================
# MSE <= 2*sigma^2/(m h^2) + C_b^2 h^4  ->  h* ~ (sigma^2/(m C_b^2))^(1/6)

function estimate_sigma2(samples_plus, samples_minus):
    # empirical variance of central-diff numerator over m pairs
    diffs = [ s_plus[i] - s_minus[i] for i in 1..m ]
    return var(diffs) / 2.0

function estimate_Cb(pilot_vals, steps):
    # crude curvature proxy: max absolute third finite-difference over a small grid of h
    # pilot_vals[h] = central difference at step h
    Cb = 0.0
    for j in 2..len(steps)-1:
        # second diff wrt h approximates leading bias coefficient
        Cb = max(Cb, abs(pilot_vals[j-1] - 2*pilot_vals[j] + pilot_vals[j+1]) / (steps[j]-steps[j-1])^2)
    return max(Cb, 1e-12)

function fd_step_size_policy(m, sigma2_hat, Cb_hat):
    h_star = (sigma2_hat / (m * Cb_hat^2))^(1/6)
    return max(h_star, 1e-8)            # floor for numerical stability

# === Dependence-aware bootstrap CIs ======================================
function block_length(n_eff, method="n_pow_third", c=1.0):
    if method == "n_pow_third":
        return max(2, floor(c * n_eff^(1/3)))
    else:
        return max(2, floor(c * n_eff^(1/3)))

function bootstrap_CI_grad(grad_series, alpha=0.05, method="cbb"):
    n = length(grad_series)
    b = block_length(n_eff)
    B = 1000
    draws = []
    for bidx in 1..B:
        if method == "cbb":
            start = randint(0, n-1)
            # wrap-around circular blocks
            idx = concatenate_blocks(start, b, n)[:n]
        else:
            idx = moving_blocks_indices(n, b)
        draws.append( mean( grad_series[idx] ) )
    q_lo = quantile(draws, alpha/2); q_hi = quantile(draws, 1 - alpha/2)
```

```
    return {"ci":[q_lo, q_hi], "b": b, "B": B}

# === End-to-end update-reselect (per estimation tick) ========================
function update_reselect(gamma, G, hatS, grad_oracle, theta, m_pairs=16, fd_steps=[1e-3,2e-3,4e-3]):
    maybe_freeze_updates()
    if freeze.active:
        log_json({"evt":"grad_skip_freeze", "window": window_id})
        return {"frozen": True}
    # 1) plane select (with degeneracy guard)
    sel = plane_select_kernel(gamma, G, hatS)
    # 2) gradient estimate: score if available; else FD with h*
    if grad_oracle.has_score:
        g_hat = grad_oracle.score(theta)             # unbiased; dependence handled in CI
        series = grad_oracle.series()                # time series for CI
        ci = bootstrap_CI_grad(series)
        method = "score"
    else:
        # pilot curvature and sigma
        pilot = {}
        for h in fd_steps:
            pilot[h] = grad_oracle.central_diff(theta, h, m=m_pairs)
        Cb_hat = estimate_Cb([pilot[h] for h in fd_steps], fd_steps)
        # variance proxy from smallest h
        sigma2_hat = estimate_sigma2( grad_oracle.samples_plus(h=fd_steps[0]),
                                      grad_oracle.samples_minus(h=fd_steps[0]) )
        h_star = fd_step_size_policy(m_pairs, sigma2_hat, Cb_hat)
        g_hat  = grad_oracle.central_diff(theta, h_star, m=m_pairs)
        series = grad_oracle.series_h(h_star)        # time series of FD grads for CI
        ci = bootstrap_CI_grad(series)
        method = "fd"
    # 3) log everything
    log_json({"evt":"grad_update",
              "window": window_id,
              "method": method,
              "g_hat": g_hat,
              "ci": ci["ci"], "b_block": ci["b"], "B_boot": ci["B"],
              "k_mode": sel["k"], "gap": float(sel["gap"]),
              "angles_L": sel["angles_L"], "angles_R": sel["angles_R"]})
    return {"plane": sel, "grad": g_hat, "ci": ci, "frozen": False}
```

Guards and good practice. - **Chatter audit.** Always log principal angles between successive planes; if the max angle exceeds a small threshold repeatedly while not in a frontier tube, raise a degeneracy warning and keep $k{=}4$ for a hold interval. - **Spectral gap hygiene.** If the top two singular values are within machine epsilon of each other, skip plane flips even if the argmax toggles sign; the J-basis is orientation-free. - **FD sanity.** Refuse $h^\star$ if the effective sample size of the FD numerator (ESS $= 1/\sum w^2$ for importance reweighting, if used) drops below a floor; increase m or take a larger h. - **CIs.** Report block length b and the bootstrap flavor (CBB/MBB) alongside the interval; this satisfies the acceptance checks in Section 9 as well.

Mini unit tests. 1) **Skew toy:** $A = \begin{bmatrix} 0 & \lambda \\ -\lambda & 0 \end{bmatrix} \oplus 0$ with random P_U, P_V. Expect $k = 2$, $\sigma_1 = \lambda$, angles $= \{0, 0\}$ across ticks. 2) **Degenerate case:** A with $\sigma_1 \approx \sigma_2$. Ensure k-escalation triggers and principal-angle drift stays bounded when you add a small perturbation. 3) **FD MSE curve:** For a known cubic $T(\theta)$, verify empirical MSE vs. h follows $ah^{-2} + bh^4$ and the measured minimum is near $h^\star$.

Uses: (C1) stratified smoothness & frontier tubes; (C3) $\tau_{\min}$; plane degeneracy threshold ε_{gap}; dependence-aware CI policy $b \sim n^{1/3}$; J-basis export to $\mathbb{A}_i$ (Def. 6).

8 Advanced Algebraic and Categorical Structures

We pin down the algebraic/categorical layer with an associative, commutative quasi-shuffle built *algebra-first*, and lift Fisher-metric control to categorical Lipschitz bounds. New here: (i) the letter-merge $\odot$ is defined by an ambient associative product (Wick/Ito/chaos) and whitening is treated as a change of basis; (ii) an *Invariant whitening* assumption ensures symmetry equivariance; (iii) a *Pushforward Lipschitz* result justifies FIM-norm non-expansiveness for Dirac/Hopf projections; (iv) monoidal submultiplicativity for functor Lipschitz constants.

8.1 Quasi-shuffle Hopf algebra by algebra-first merge

Alphabet and degrees. Let $\mathcal{A} = \{a_{p,q}^{(m)}\}$ be the centered, whitened dictionary of sufficient statistics (Section 5); $\deg(a_{p,q}^{(m)}) = (p, q)$, total degree $|a| = p + q$. Let $H = \mathbb{K}\langle\mathcal{A}\rangle$ be the tensor algebra of words with empty word 1 and ground field $\mathbb{K}$ of characteristic 0.

Assumption 3 (Invariant whitening)**.** *[Inf/Alg]* The inner product used to whiten $\mathcal{A}$ is invariant under the symmetry group acting on Γ and G (e.g., orthogonal frames on Γ, graph isomorphisms on G). Concretely, whitening is defined by a symmetry-invariant covariance (orbit-average or group-invariant frame), so the change-of-basis maps commute with group actions.

Uses: (C5) Fisher inner product at P_0; symmetry group; whitening map.

Algebra-first letter merge. Choose an ambient *associative, commutative* product $\diamond$ on centered variables (e.g., Wick/Ito/chaos on the EF statistics). Define the letter-merge on $\mathcal{A}$ by structure constants

$$a_{p,q}^{(m)} \odot a_{r,s}^{(m')} \;:=\; \sum_{m''} c^{m''}_{(m,p,q),(m',r,s)}\, a^{(m'')}_{p+r,q+s},$$

where $c^{m''}$ are the expansion coefficients of $a_{p,q}^{(m)} \diamond a_{r,s}^{(m')}$ in the *ambient* basis, and *then* whiten by an invertible linear map W (change of basis).

Proposition 4 (Associativity preserved under change of basis)**.** [Alg] *If $\odot$ is induced by an associative, commutative bilinear map $\diamond$ on the ambient letters and W is invertible, then the structure constants in the whitened basis still define a commutative, associative $\odot$.*

Proof sketch. Let B be the ambient basis and $\tilde{B} = W(B)$ the whitened one. Structure tensors transform by $C \mapsto (W^{-1})^{\otimes 1} C\, W^{\otimes 2}$. Associativity $C(C(\cdot,\cdot),\cdot) = C(\cdot, C(\cdot,\cdot))$ is a tensor identity preserved by invertible congruences. □

Uses: Ambient associative $\diamond$; invertible W.

Quasi-shuffle (*stuffle*) product. Define $u \uplus v$ recursively for words u, v and letters a, b:

$$1 \uplus w = w \uplus 1 = w, \qquad (ua) \uplus (vb) = (u \uplus vb)\, a + (ua \uplus v)\, b + (u \uplus v)\, (a \odot b).$$

By Proposition 4, $(H, \uplus)$ is a commutative, associative, graded algebra.

Coalgebra and antipode. With deconcatenation $\Delta(w) = \sum_{uv=w} u \otimes v$ and counit $\epsilon(1) = 1$, $(H, \uplus, \Delta, \epsilon)$ is graded connected and admits the standard antipode (Takeuchi), hence a commutative, noncocommutative Hopf algebra.

Proposition 5 (Symmetry equivariance)**.** [Alg/Rep] *Under Assumption 3, group actions on $\mathcal{A}$ act by Hopf automorphisms: the action commutes with $\odot$, $\uplus$, and Δ.*

Proof sketch. Invariance makes W and the structure constants $c^{m''}$ equivariant; the recursion and deconcatenation are natural. □

Uses: Assumption 3; ambient product equivariance.

8.2 Characters, cumulants, and the degree-3 bridge

Characters and cumulants. Any law P induces a character $\chi_P : H \to \mathbb{K}$ by $\chi_P(1) = 1$, $\chi_P(uv) = \chi_P(u)\chi_P(v)$, and $\chi_P(a) = \mathbb{E}_P[T_a]$. The cumulant character is $\kappa_P := \log_* \chi_P = \sum_{n\geq 1} \frac{(-1)^{n-1}}{n}(\chi_P - \epsilon)^{*n}$, which is *primitive*: $\Delta(\kappa_P) = \kappa_P \otimes 1 + 1 \otimes \kappa_P$.

Worked order-3 example and the $\kappa_{2,1}$ split. Let $x := a^{(i)}_{1,0}$, $y := a^{(j)}_{1,0}$, $z := a^{(\ell)}_{0,1}$. The degree-$(2,1)$ Eulerian idempotent $E_{2,1}$ evaluates to the mixed cumulant $\kappa_{2,1}$ under any character:

$$(\chi_P \otimes \chi_P)\,\Delta\big(E_{2,1}\big) = \kappa_{2,1} \otimes 1 + 1 \otimes \kappa_{2,1} + \kappa_{1,1} \otimes \kappa_{1,0} + \kappa_{2,0} \otimes \kappa_{0,1} + \kappa_{1,0} \otimes \kappa_{1,1}.$$

This justifies keeping $\kappa_{2,1}$ and its degree-≤ 2 factors at order $K = 3$.

8.3 Hopf-canonical truncation and residual

Let $H^{(\leq K)}$ be the span of words with $|w| \leq K$ and define the orthogonal projector

$$\Pi_{\leq K} : H \to H^{(\leq K)}, \qquad \Pi_{\leq K}(w) = \begin{cases} w, & |w| \leq K, \\ 0, & |w| > K. \end{cases}$$

The closure residual in Section 4 is exactly $\|(I - \Pi_{\leq K})\hat{S}\|$ in the whitened inner product.

8.4 Functor Lipschitz constants and non-expansiveness in FIM

Objects and metric. For an instance X, let $\mathbf{k}_K(X)$ be the stacked whitened cumulant vector up to degree K and

$$d_K(X, X') := \|\mathbf{k}_K(X) - \mathbf{k}_K(X')\|_{\mathrm{FIM}^{-1}}.$$

Proposition 6 (Pushforward Lipschitz)**.** [Comp/Inf] *Let $T : X \to X$ be L-Lipschitz in d_X and each statistic in the dictionary be C-Lipschitz on X. In an EF neighborhood of P_0,*

$$\|\mathbf{k}_K(T_\# P_\theta) - \mathbf{k}_K(P_\theta)\|_{\mathrm{FIM}^{-1}} \;\leq\; L\,C\,\|\theta\| \;+\; O(\|\theta\|^2).$$

Proof sketch. Write $\kappa(\theta) = \nabla\psi(\theta)$. For any letter a, $\big|\mathbb{E}_{P_\theta}[T_a \circ T] - \mathbb{E}_{P_\theta}[T_a]\big| \leq LC\,W_1(P_\theta, P_0) = O(\|\theta\|)$; multilinearity and the EF expansion then give the bound in the Fisher norm. □

Uses: (C4) window/EFradius; (C5) F, dictionary Lipschitz C, map Lipschitz L.

Lemma 12 (Non-expansive projections)**.** [Comp/Inf/Ctrl] *Dirac Π (Section 6) and Hopf $\Pi_{\leq K}$ satisfy*

$$\|\mathbf{k}_K(\Pi X) - \mathbf{k}_K(\Pi X')\|_{\mathrm{FIM}^{-1}} \;\leq\; \|\mathbf{k}_K(X) - \mathbf{k}_K(X')\|_{\mathrm{FIM}^{-1}},$$

and $\|\mathbf{k}_K(\Pi X)\|_{\mathrm{FIM}^{-1}} \leq \|\mathbf{k}_K(X)\|_{\mathrm{FIM}^{-1}}$.

Proof sketch. Both Π and $\Pi_{\leq K}$ are orthogonal projections (energetic and Hopf/whitened, respectively), hence $L = 1$. Each dictionary element is $C \leq 1$-Lipschitz w.r.t. the induced Hilbert norm. Apply Proposition 6 and take $X' = 0$ for the one-argument bound. □

Uses: Proposition 6; energetic orthogonality; whitened orthogonality; (C5) Fisher norm.

Theorem 4 (Monoidal submultiplicativity)**.** [Comp] *If F, G are C_F- and C_G-Lipschitz strong monoidal functors between the K-cumulant categories, then $(F \otimes G)$ is $C_F C_G$-Lipschitz:*

$$d_K\big((F \otimes G)(X), (F \otimes G)(X')\big) \;\leq\; C_F C_G\, d_K(X, X').$$

Proof. Block-diagonal Fisher metrics under independence and the direct sum of cumulants yield the product bound. □

Uses: Block-diagonal Fisher metric; functorial monoidality.

Proposition 7 (Local/Orlicz Lipschitz)**.** *On any sublevel set $\{x : \Phi(x) \le R\}$ where each letter in $\mathcal{T}_K$ is $C(R)$-Lipschitz, the pushforward bound of Prop. 6 holds with $C = C(R)$. If letters have sub-Gaussian tails, then replacing Lipschitz norms by Orlicz ψ_2 norms yields the same first-order bound in expectation.*

Remark 6 (Equivariance error from orbit-averaging). Let $\widehat{\Sigma}$ be the Monte Carlo orbit-averaged covariance and Σ_G the exact group integral. If $\|\widehat{\Sigma} - \Sigma_G\| \le \epsilon$, then the quasi-shuffle structure constants (hence Lipschitz constants) perturb by $O(\epsilon)$. *Report:* log the observed ϵ per window.

Acceptance checks for Section 8.

1. The letter-merge $\odot$ is *algebra-first* with associativity guaranteed (Prop. 4); projection is a basis change, not a definition.
2. Symmetry claims cite Assumption 3 and Prop. 5.
3. Non-expansiveness in the FIM metric points to the *Pushforward Lipschitz* Proposition 6 and then Lemma 12.
4. Theorem 4 is stated and used wherever monoidal products appear.

Numerical API (Hopf algebra, invariant whitening, FIM Lipschitz tests)

```
# === Inputs (conceptual) ==========================================================
Dictionary:
  A_letters = { a_{p,q}^{(m)} : total degree |a|<=K }     # centered sufficient-statistic letters
  index(a) <-> integer id                                  # bijection letter <-> id

Ambient product:
  ambient_prod(i,j) -> list[(k, coeff)]                    # coefficients in ambient (pre-whitened) basis
                                                           # e.g., Wick/Ito/chaos product on centered vars

Whitening (Assumption 8.1):
  # symmetry-invariant inner product at P0:
  #   W: ambient -> whitened basis, with  <Wx,Wy> = x^T Sigma^{-1} y   (Sigma from FIM/covariance at P0)
  Sigma = fisher_covariance_at_P0(A_letters)               # SPD; symmetry-invariant construction
  W = inv(chol(Sigma))                                     # whiten: a_tilde = W * a_ambient

Samples / cumulants:
  T_stream  # time/windowed samples of letter evaluations T_a(x)
  F = FIM_at_P0_whitened()                                 # equals I_d if fully whitened

Logging:
  log_json(dict)
  window_id

# === Build letter-merge structure constants (algebra-first) ==================
# 1) Ambient structure tensor C_ambient from ambient_prod
function build_C_ambient():
    d = len(A_letters)
    # store as sparse: dict[(i,j)] -> list[(k, c)]
    C = dict()
    for i in 1..d:
        for j in 1..d:
            C[(i,j)] = ambient_prod(i,j)                   # list of (k, c_ambient)
    return C

```

```
# 2) Change of basis to whitened constants via associativity-preserving congruence
#    If A_tilde = W A_amb, then structure tensor transforms by   C_tilde = W * C_ambient * (W^{-1} otimes-- W^{-1})
#    Implement as: coefficients in whitened basis via least-squares with orthonormal W (Cholesky choice makes W^T W
function whiten_constants(C_ambient, W):
    # For each pair (i,j), transform the coefficient vector in ambient coords to whitened coords.
    # Practical: apply W to the RHS basis and W^{-1} to inputs; since W is invertible, we can precompute Winv = inv
    Winv = inv(W)
    C_tilde = dict()
    for (i,j), terms in C_ambient.items():
        # Build coefficient vector c_amb in ambient coordinates
        c_amb = zeros(d)
        for (k, c) in terms: c_amb[k] += c
        # Transform: c_tilde = W * c_amb, inputs indices mapped by Winv (absorbed into basis reindexing)
        c_tilde = W @ c_amb
        # Sparsify
        out = []
        for k in 1..d:
            if abs(c_tilde[k]) > 1e-12: out.append((k, c_tilde[k]))
        C_tilde[(i,j)] = out
    return C_tilde

# 3) Associativity spot-check (sanity):  (a \odot b) \odot c  ==  a \odot (b \odot c)
function check_associativity(C_tilde, trials=256, tol=1e-8):
    d = len(A_letters)
    for t in 1..trials:
        i,j,k = randint(1,d), randint(1,d), randint(1,d)
        # left: L = C(C(e_i,e_j), e_k)
        L = zeros(d)
        for (h, cij) in C_tilde[(i,j)]:
            for (r, chr) in C_tilde[(h,k)]:
                L[r] += cij * chr
        # right: R = C(e_i, C(e_j, e_k))
        R = zeros(d)
        for (h, cjk) in C_tilde[(j,k)]:
            for (r, cir) in C_tilde[(i,h)]:
                R[r] += cjk * cir
        if norm(L-R, inf) > tol:
            return {"assoc_ok": False, "i":i,"j":j,"k":k,"diff": norm(L-R,inf)}
    return {"assoc_ok": True}

# === Quasi-shuffle (stuffle) product on words with letter-merge C_tilde ======
# Words represented as lists of letter ids; use memoization for speed.
memo = dict()
function uplus(u, v, C_tilde):
    key=(tuple(u),tuple(v))
    if key in memo: return memo[key]
    if len(u)==0: return {v:1.0}
    if len(v)==0: return {u:1.0}
    u1, a = u[:-1], u[-1]
    v1, b = v[:-1], v[-1]
    # (u\uplusvb) * a
    T1 = { w+[a]: c for (w,c) in uplus(u, v1, C_tilde).items() }
    # (ua\uplusv) * b
    T2 = { w+[b]: c for (w,c) in uplus(u1, v, C_tilde).items() }
    # (u\uplusv) * (a\odotb)
    T3 = dict()
    for (k, cab) in C_tilde[(a,b)]:
        for (w,c) in uplus(u1, v1, C_tilde).items():
            w2 = w+[k]; T3[w2] = T3.get(w2,0.0) + c*cab
```

```
    # sum
    out = T1
    for (w,c) in T2.items(): out[w] = out.get(w,0.0)+c
    for (w,c) in T3.items(): out[w] = out.get(w,0.0)+c
    memo[key]=out; return out

# === Symmetry equivariance audit (Assumption 8.1) ============================
# Given group actions G on letters (permutation/orthogonal), verify invariance of C_tilde.
function check_equivariance(C_tilde, actions):
    # actions: list of group elements acting as permutation/orhogonal maps P_g on indices
    for g in actions:
        P = action_matrix(g)            # dxd, orthogonal/permutation
        for (i,j), terms in C_tilde.items():
            # pushforward both inputs by g and compare pushed outputs
            i2, j2 = P@e_i, P@e_j       # conceptual; implement as index mapping
            lhs = push_constants(C_tilde, i2, j2)
            rhs = push_forward_output(terms, P)   # apply P to output index
            if not approx_equal(lhs, rhs, tol=1e-10):
                return {"equiv_ok": False, "g": g, "ij": (i,j)}
    return {"equiv_ok": True}

# === Empirical FIM-norm contraction tests for projections ====================
# Build cumulant vector k_K from T_stream (order <= K); then compare norms before/after projections.
function cumulant_vector_K(T_stream, K):
    # compute centered moments ->' cumulants (order <=3 typically); return flattened vector in whitened basis
    return k_vec

function FIM_norm(vec, F):
    # vec is already whitened; with perfect whitening F\approxI, but allow general F:
    return sqrt( vec^T * inv(F) * vec )

function test_projection_contraction(Pi_op, PiK_op, T_stream, F, K):
    k_full = cumulant_vector_K(T_stream, K)
    # Dirac projection \Pi (acts on trajectories/mechanics ->' induces pushforward of T_stream)
    T_proj = Pi_op.apply_to_stream(T_stream)
    k_dirac = cumulant_vector_K(T_proj, K)
    # Hopf truncation \Pi_{<=K} (acts directly in algebra; numerically: zero out >K words)
    k_hopf = PiK_op.apply_to_cumulants(k_full)      # or recompute on truncated dictionary

    n_full  = FIM_norm(k_full,  F)
    n_dirac = FIM_norm(k_dirac, F)
    n_hopf  = FIM_norm(k_hopf,  F)

    log_json({"evt":"fim_contraction_test", "window": window_id,
              "||k||": n_full, "||Pi k||": n_dirac, "||Pi<=K k||": n_hopf,
              "dirac_ok": (n_dirac <= n_full + 1e-10),
              "hopf_ok":  (n_hopf  <= n_full + 1e-10)})
    return {"dirac_ok": n_dirac<=n_full+1e-10, "hopf_ok": n_hopf<=n_full+1e-10,
            "n_full": n_full, "n_dirac": n_dirac, "n_hopf": n_hopf}

# === Pushforward Lipschitz spot-check (Prop. 8.5) ============================
# Given a Lipschitz map T (geometry layer) and small EF tilt theta, check:
#   ||k_K(T_# P_theta) - k_K(P_theta)||_{F^{-1}} <= L*C*||theta|| + O(||theta||^2)
function pushforward_Lip_check(T_map, theta, L_est, C_est, T_stream_base, F, K):
    # importance-reweight T_stream_base to P_theta if needed (small tilt); or draw fresh samples
    k_theta   = cumulant_vector_K( reweight(T_stream_base, theta), K )
    k_theta_T = cumulant_vector_K( reweight(T_map.apply_to_stream(T_stream_base), theta), K )
    diff = FIM_norm(k_theta_T - k_theta, F)
    rhs  = L_est * C_est * norm(theta)
```

```
    log_json({"evt":"push_Lip_check", "window": window_id, "lhs": diff, "rhs": rhs,
              "ok": diff <= 1.2*rhs})     # allow small slack for finite-sample noise
    return {"ok": diff <= 1.2*rhs, "lhs": diff, "rhs": rhs}

# === End-to-end builder (one call) ==========================================
function build_quasi_shuffle_runtime(K, actions=None):
    C_amb = build_C_ambient()
    C = whiten_constants(C_amb, W)
    assoc = check_associativity(C)
    if not assoc["assoc_ok"]:
        log_json({"evt":"assoc_fail", "detail": assoc}); raise "Associativity spot-check failed"
    if actions is not None:
        eq = check_equivariance(C, actions)
        log_json({"evt":"equiv_check", "ok": eq["equiv_ok"]})
    return {"C_tilde": C, "uplus": (u,v) -> uplus(u,v,C)}
```

Guards and good practice. - **Associativity audit.** Keep the spot-check enabled in CI; any failure means your ambient product or whitening map isn't consistent. - **Invariant whitening.** If the symmetry group is large, approximate the orbit-average covariance by Monte Carlo over group elements; log the number of samples and the max deviation from exact invariance. - **Sparsity.** Store C_{tilde} as sparse maps; quasi-shuffle growth is combinatorial — cut off strictly at degree K and cache results per word length. - **Projection tests.** Run the FIM-contraction test for both Π and $\Pi_{\leq K}$ per window; keep failures visible in the auto-report. - **Pushforward Lipschitz.** Use the same small-tilt reweighting machinery as Section 5's L_ψ estimator; ensure effective sample size is healthy (e.g., ESS $> 0.2n$).

Mini unit tests. 1) **Ambient ->¦ whitened consistency:** choose a quadratic-Gaussian ambient where Wick coefficients are analytic; verify associativity passes for random triples. 2) **Equivariance:** apply a random orthogonal frame change on Γ; check that transformed constants match via the action test. 3) **Non-expansiveness (empirical):** simulate data, compute $\|\mathbf{k}_K\|_{\mathrm{FIM}^{-1}}$ before/after Π and $\Pi_{\leq K}$; both should not increase beyond numerical tolerance.

Uses: Invariant whitening (Assumption 3); Prop. 4, Prop. 6, Lem. 12; FIM F and degree cap K.

9 Computational Statistics and Practicalities

We fix a uniform, dependence-aware inference policy for all reported quantities: the mixed-cumulant vector $\mathbf{k}_K = (\kappa_{p,q}^{(m)})_{1\leq p+q\leq K}$, the gradient vectors used for shape derivatives, and the spectral gap $\Delta\sigma := \sigma_1 - \sigma_2$ driving plane selection. All norms are Fisher-induced (Theorem 2) unless noted.

9.1 Data model, estimands, and dependence

Let $\{Z_t\}_{t=1}^n$ be the windowed time series of centered, whitened test functionals (Section 3) from which we compute moments/cumulants up to order $K \in \{2, 3\}$. Gradient estimators and SVD gaps are functions of the same stream.

Assumption 4 (Mixing and moments)**.** *[Stoch]* Within a window, $\{Z_t\}$ is strictly stationary and *strongly mixing* with coefficients $\alpha(k)$ satisfying $\sum_{k\geq 1} \alpha(k)^{\delta/(2+\delta)} < \infty$ for some $\delta > 0$, and $\mathbb{E}\|Z_t\|^{4+\delta} < \infty$. (Windowwise analysis uses the same assumptions per window.)

Uses: Window size τ; effective n per window.

9.2 HAC covariance and Wald CIs (default for smooth functionals)

Let $\widehat{\mathbf{k}}_K$ be the estimator of $\mathbf{k}_K$ (streaming U-stats for order ≤ 3, with sketching optional for order 3). Let $\{\widehat{\psi}_t\}$ be the associated influence-function process (componentwise).

Lemma 13 (HAC CLT and consistency)**.** [Stoch/Comp] *Under Assumption 4,* $\sqrt{n}\big(\widehat{\mathbf{k}}_K - \mathbf{k}_K\big) \xrightarrow{d} \mathcal{N}(0, \Sigma_K)$*, where* Σ_K *is the long-run covariance. Let* $\widehat{\Sigma}_K$ *be a HAC estimator with a positive, compact-support taper (e.g., Bartlett) and bandwidth* $b_{\mathrm{HAC}} = \lfloor c\, n^{1/3} \rfloor$ *for fixed* $c > 0$*. Then* $\widehat{\Sigma}_K \xrightarrow{p} \Sigma_K$ *and Wald CIs using* $\widehat{\Sigma}_K$ *are asymptotically valid.*

Uses: $b_{\mathrm{HAC}} \sim n^{1/3}$*; taper choice; moment/mixing envelope.*

Remark 7 (Reporting policy for HAC). Every table/figure that includes $\widehat{\mathbf{k}}_K$ must print: the taper, b_{HAC}, and the FIM norm used to scale Wald ellipsoids.

Uses: Fisher F; user-chosen c.

9.3 Block bootstrap CIs (default for non-smooth or finite-difference estimators)

For gradients (score/FD) and the spectral gap $\Delta\sigma$, we use block bootstrap CIs.

Proposition 3 (MBB/CBB validity and block-length rule)**.** [Stoch/Comp] *Let* $\widehat{\theta}$ *be any smooth functional of* $\{Z_t\}$ *or a finite-difference estimator with* $h \to 0$ *and* $mh^2 \to \infty$*. The moving (MBB) or circular (CBB) block bootstrap with block length* $b = \lfloor c\, n^{1/3} \rfloor$ *yields consistent percentile or studentized CIs for* $\widehat{\theta}$ *under Assumption 4. A data-driven choice of* c *by pilot MSE minimization (variance + squared-bias proxy) is admissible and recommended.*

Uses: $b \sim n^{1/3}$*; pilot MSE; FD step* h *(§7.2).*

Remark 8 (Which CI where). Use HAC/Wald for $\widehat{\mathbf{k}}_K$ and other smooth functionals; use block bootstrap for: (i) finite-difference gradients, (ii) SVD gaps and k-plane engagement rates, (iii) any estimator involving hard thresholds or projections. Always print the CI type and (b, B) in logs and captions.

Uses: Estimator smoothness; bootstrap design (b, B).

9.4 Simultaneous inference and FDR control

Let $\{\widehat{T}_j\}_{j=1}^m$ be the family of tested entries (e.g., coordinates of $\mathbf{k}_K$). Let $\widehat{p}_j$ be marginal p-values (HAC-based z-tests or bootstrap studentized).

Lemma 14 (BH/BY control)**.** [Stoch/Comp/Inf] *At target level* $q \in (0, 1)$*: (i) If the p-values are PRDS (positively dependent), the Benjamini–Hochberg procedure controls FDR at* q*. (ii) Without dependence assumptions, Benjamini–Yekutieli controls FDR at* q *after scaling* $q \leftarrow q/\sum_{i=1}^m 1/i$*. A hierarchical variant—BH/BY within groups (by bidegree* (p, q) *and stratum), then BH on group-level p—is admissible and recommended.*

Uses: Group partition by bidegree/stratum; chosen q*.*

Remark 9 (Symmetry projection). Enforce bidegree symmetries and invariances by projecting $\widehat{\mathbf{k}}_K$ onto the symmetry-fixed subspace (orthogonal in the whitened inner product) before testing; this reduces multiplicity and increases power without inflating type-I error.

Uses: Invariant whitening (Assumption 3); orthogonal projector.

9.5 Gradients and finite differences: CIs and step choice

For gradients, apply Proposition 3 (score identity and FD MSE) and use the bootstrap for CIs.

Lemma 15 (CI policy for gradients)**.** [Stoch/Inf] *Compute* $h^\star \asymp (\sigma^2/(mC_b^2))^{1/6}$ *and form CIs by CBB/MBB with* $b \sim n^{1/3}$ *on the gradient time series (score or FD at* $h^\star$*). Report* $(h^\star, b, B)$ *and whether* frontier freeze *was active.*

Uses: Proposition 3; block bootstrap; frontier freeze flag.

9.6 SVD gaps and k-plane engagement: CIs and certification

Lemma 16 (Gap certification)**.** [Comp/Ctrl] *Estimate $\Delta\sigma_t$ per window and report a bootstrap CI; additionally, report principal-angle bands from Davis–Kahan/Wedin bounds using the observed perturbation proxy. Trigger k-plane escalation when $\widehat{\Delta}\sigma_t < \varepsilon_{\text{gap}}$ or its CI contains 0.*

Uses: Degeneracy threshold ε_{gap}; bootstrap CI; perturbation proxy.

9.7 Sketching and variance propagation for order-3

When tensor sketching (CountSketch/Hutch++) is used for third-order moments:

Lemma 17 (Sketch error propagation)**.** [Comp] *With sketch size $s = O(\epsilon^{-2} \log d)$, the sketched third-order moment yields a cumulant error $\|\widehat{\kappa}_{2,1} - \kappa_{2,1}\| = O_{\mathbb{P}}(\epsilon)$ for fixed n; the HAC/BB procedures apply with an added $O(\epsilon)$ bias term that vanishes as $s \to \infty$.*

Uses: Sketch size s; target ϵ.

9.8 Reporting template (uniform across outputs)

Every reported estimate *must* include: (i) CI type (HAC vs. MBB/CBB), taper and b_{HAC} or (b, B); (ii) FIM norm used to scale ellipsoids; (iii) for gradients: $h^\star$ and freeze flag; (iv) for SVD gaps: k-plane mode, $\widehat{\Delta}\sigma$, its CI, and principal-angle bands; (v) for multiple tests: the FDR procedure (BH/BY), group partition, and target q.

Acceptance checks for Section 9.

1. HAC objects print taper and b_{HAC}; bootstrap objects print (b, B) and method (MBB/CBB).
2. Gradient outputs print $h^\star$ and the frontier freeze status.
3. SVD reports include gap CIs and principal-angle bands; k-plane engagement is logged.
4. Any multiple-testing claim includes BH/BY choice, group structure, and target q.

Numerical API (HAC, block bootstrap, FDR reporting)

```
# Inputs
stream:      time series or windowed array of estimates/influence values
est_name:    string tag (e.g., "kappa_21", "grad_theta", "gap_svd")
n_eff:       effective sample size (# points in window)
alpha:       0.05 (default)
q_fdr:       target FDR for families of tests (e.g., entries of k_K)
F:           Fisher matrix for FIM norm (if needed)

# HAC covariance for smooth vector estimators ---------------------------------
function hac_cov(stream_vec, c=1.0, taper="bartlett"):
    n = len(stream_vec)
    b = max(2, floor(c * n**(1/3)))
    S0 = empirical_cov(stream_vec)                          # lag 0
    S = S0
    for h in 1..b:
        w = 1 - h/(b+1) if taper=="bartlett" else kernel_weight(h,b,taper)
        Gh = lagged_cov(stream_vec, h)                      # sample lag-h cov
        S += w * (Gh + Gh.T)
    log_json({"evt":"hac", "name": est_name, "b_HAC": b, "taper": taper})
    return S
```

```
function wald_ci(point, cov, alpha=0.05):
    se = sqrt(diag(cov))
    z = normal_quantile(1 - alpha/2)
    lo = point - z * se; hi = point + z * se
    return {"ci_lo": lo, "ci_hi": hi, "se": se, "z": z}

# Block bootstrap CI (MBB/CBB) ------------------------------------------------
function block_length(n_eff, c=1.0): return max(2, floor(c * n_eff**(1/3)))

function bootstrap_ci(series, alpha=0.05, method="cbb", B=1000, c=1.0):
    n = length(series); b = block_length(n, c)
    draws = []
    for bidx in 1..B:
        idx = sample_blocks(n, b, method)                  # moving or circular
        draws.append( mean(series[idx]) )
    q_lo = quantile(draws, alpha/2); q_hi = quantile(draws, 1 - alpha/2)
    log_json({"evt":"bootstrap", "name": est_name, "method": method, "b": b, "B": B})
    return {"ci":[q_lo, q_hi], "b": b, "B": B}

# Gradient CI wrapper ---------------------------------------------------------
function grad_ci(grad_series, h_star, alpha=0.05, method="cbb"):
    ci = bootstrap_ci(grad_series, alpha=alpha, method=method)
    log_json({"evt":"grad_ci", "name": est_name, "h_star": h_star,
              "ci": ci["ci"], "b": ci["b"], "B": ci["B"]})
    return ci

# SVD gap CI and principal-angle band -----------------------------------------
function gap_ci(gap_series, angles_series=None, alpha=0.05):
    ci = bootstrap_ci(gap_series, alpha=alpha, method="cbb")
    out = {"gap_ci": ci, "angles": None}
    if angles_series is not None:
        out["angles"] = {"max_angle_ci": bootstrap_ci(angles_series, alpha=alpha)}
    log_json({"evt":"gap_ci", "name": est_name, "ci": ci["ci"]})
    return out

# Multiple testing: BH/BY -----------------------------------------------------
function fdr_bh(pvals, q=0.1):
    m = len(pvals); order = argsort(pvals)
    thresh = [ (k/m) * q for k in 1..m ]
    k_star = max({k: pvals[order[k]] <= thresh[k]} \\\\cup {0})
    reject = zeros(m, dtype=bool); reject[order[:k_star]] = True
    log_json({"evt":"fdr_bh", "m": m, "q": q, "k_star": int(k_star)})
    return reject

function fdr_by(pvals, q=0.1):
    m = len(pvals); Hm = sum_{i=1}^m (1/i)
    return fdr_bh(pvals, q=q/Hm)

function hierarchical_fdr(groups):
    # groups: dict[group_id] -> list of p-values
    rejects = dict()
    for g, pv in groups.items():
        rejects[g] = fdr_bh(pv, q=q_fdr)                   # or BY per policy
    # combine group-level with a BH on min-p per group if needed
    return rejects

# Uniform reporting helper ----------------------------------------------------
function report_estimate(name, point, ci_info, meta):
```

```
    log_json({"evt":"report", "name": name,
              "point": point, "ci": ci_info,
              "meta": meta})    # include: HAC/BB settings, FIM norm, h_star, freeze flag, k-plane, etc.
```

10 Benchmarks and Application Domains

This section provides *executable* protocols and an *auto-report* that compiles the exact quantities our guarantees require (ISS/PR, frontier terms, cumulant CIs/FDR, plane degeneracy metrics). All protocols reuse the same pipeline (windows, cumulants, SVD planes, Dirac projection, PR checks, DRO closure) and differ only by environment adapters. The auto-report consumes windowed JSON logs produced by Section Section 6–9.

10.1 Reproducibility pack (common to all domains)

Artifacts. (i) Raw/processed data (or generator seeds); (ii) YAML configs; (iii) source with fixed versions; (iv) lockfile; (v) seeds; (vi) per-window JSON logs (metrics, CI, PR margins); (vii) notebooks to render Tables A–D and the figures below.

Config schema (minimal). Keys controlling windows, cumulant order K, bootstrap and HAC settings, Oustaloup band/order, degeneracy threshold ε_{gap}, and DRO radius policy (c, τ) are identical across domains.

```
windows:{size: 2048, stride: 512} stats:{order_K:[2,3], hac_c:1.0, bb_B:1000} svd:{gap_thresh:
0.1} control:{oustaloup:{N:7, band:[0.5,8.0]}, pr_margin_min:0.05} dro:{c:1.5, tau:2048}
```

10.2 Negative control: factorized ground truth

Design. Construct data with $\gamma \perp G$ by design (independent factors or blocked randomization); generate statistics with the same dictionary, whitening, and windowing as treatment runs.

Targets. For each window w:

$$\|\widehat{\kappa}_{\text{mix}}^{(w)}\|_{\text{FIM}^{-1}} \text{ CI includes } 0, \qquad \widehat{\eta}^{\star(w)} \approx 0 \quad \text{in the DRO dual.}$$

Pass/Fail. *Pass* if: (i) the HAC ellipsoid for $\mathbf{k}_K^{(w)}$ contains the origin in at least 95% of windows, *and* (ii) the median $\widehat{\eta}^{\star}$ across windows is within a small neighborhood of 0 (tunable, e.g. ≤ 0.1). Otherwise *fail*. Also report the BH/BY-adjusted discovery count; it should be 0 (up to the nominal FDR).

Uses: *(Stoch/Inf/Clos)* HAC policy (Section 9), DRO dual (Section 4), invariant whitening (Section 8).

10.3 Degeneracy stress: near-multiplicity of top singular values

Design. Create episodes where the constrained operator A_{UV} satisfies $\sigma_1 \approx \sigma_2$ for an extended interval by tuning environment parameters (e.g., symmetric contact geometry). Trigger the k-plane policy when $\Delta\sigma = \sigma_1 - \sigma_2 < \varepsilon_{\text{gap}}$.

Targets. For each window:

- bootstrap CI for $\Delta\sigma$ (CBB/MBB, $b \sim n^{1/3}$);
- principal-angle bands (Davis–Kahan/Wedin) between successive selected planes;
- chatter rate (fraction of windows with max principal angle above a small threshold while not in a frontier tube);
- safety/goal metrics (e.g., $d_{\min}$ in robotics) unchanged relative to non-degenerate runs.

Pass/Fail. *Pass* if: (i) the k-plane mode engages when $\widehat{\Delta}\sigma$ CI intersects 0; (ii) the empirical principal angles respect the certified Wedin band in all but a $\leq 5\%$ tail; (iii) chatter rate $\leq 1\%$ outside frontier tubes; (iv) no degradation in domain safety metric beyond CI overlap.

Uses: *(Ctrl/Comp)* degeneracy guard (Section 7), bootstrap CI (Section 9), frontier freeze (Section 6).

10.4 Common auto-report (Tables & Figures)

Table A: Config. Windows, K, HAC bandwidth b_{HAC}, bootstrap (b, B), controller $(N, [\omega_L, \omega_H])$, PR margin summary.

Table B: Domain metrics. Means $\pm$ CIs and effect sizes (Cliff's δ) vs. baselines.

Table C: Statistical layer. $\|\widehat{\mathbf{k}}_K\|_{\mathrm{FIM}^{-1}}$ with HAC ellipsoid, BH/BY discoveries by bidegree, negative-control checks.

Table D: Control layer. PR margin $[\min, \max]$, spectral gaps, % time in k-plane mode, frontier crossings, ISS terms:

$$C_0 N_{\mathrm{cross}}(0,t), \qquad \frac{C_0}{\tau_{\min}}\, t.$$

Figure 1. Lead-time/performance curves (or generalization curves in neuroscience).

Figure 2. Degeneracy/stability (gap CI and principal-angle bands).

Figure 3. Windowed drift: $\widehat{L}_t$, $\rho_t = c\,\widehat{L}_t\,\tau$, and $\widehat{\eta}_t^\star$ trajectories.

Acceptance checks for Section 10.

1. Negative control passes both: origin-inclusion for $\mathbf{k}_K$ and $\widehat{\eta}^\star \approx 0$.
2. Degeneracy stress satisfies: k-plane engagement on gap-CI touch, Wedin-certified principal angles, low chatter, preserved safety metric.
3. Auto-report prints HAC taper/b_{HAC}, bootstrap (b, B), FDR q, PR margin range, % time in k-plane, frontier counts, and the explicit ISS frontier terms.

Auto-report additions (Section 10).

- *Whitening leakage alarms:* log $\chi(t)$ and flag any $\chi(t) > \chi_{\max}$ with the window id.
- *DRO metric sensitivity:* compute $\|\partial_\epsilon \theta^*(0)\|$ from Prop. 6 and print as "DRO-metric sensitivity".

Numerical API (auto-report aggregation for Section 10)

```
# === Inputs ============================================================
logs:         iterable of per-window JSON events from Section~Section~6--9
domain:       {"robotics","neuroscience","finance",...}
alpha:        0.05
fdr_q:        0.1

# === Collectors ========================================================
state agg = {
  "config": {}, "windows": [],
```

```
  "hac": [], "bootstrap": [],
  "kappa": [], "kappa_fim_norm": [],
  "fdr": {"pvals_by_group": {}},
  "pr": [], "svd": [], "angles": [],
  "frontier": {"N_cross":0, "events":[]},
  "iss": [], "drift": [], "dro_dual": []
}

function ingest(logs):
    for evt in logs:
        switch evt["evt"]:
          case "hac":
            agg["hac"].append(evt)                          # name, b_HAC, taper
          case "bootstrap":
            agg["bootstrap"].append(evt)                    # name, method, b, B
          case "report":
            if "kappa" in evt["name"]:
                agg["kappa"].append(evt)
                if "fim_norm" in evt["meta"]:
                    agg["kappa_fim_norm"].append({"window":evt["meta"]["window"],
                                                  "norm":evt["meta"]["fim_norm"]})
          case "fim_contraction_test":
            agg["pr"].append({"window":evt["window"], "n_full":evt["||k||"],
                              "n_dirac":evt["||Pi k||"], "n_hopf":evt["||Pi<=K k||"]})
          case "ctrl_update":
            agg["pr"].append({"window":evt["t"], "eps":evt["pr_eps"], "cert":evt["pr_cert"]})
          case "plane_select":
            agg["svd"].append({"window":evt["window"], "sigma_top":evt["sigma_top"], "gap":evt["gap"],
                               "k_mode":evt["k_mode"], "degenerate":evt["degenerate"]})
            if evt["angles_L"] is not None:
                agg["angles"].append({"window":evt["window"], "max_angle": max(evt["angles_L"]+evt["angles_R"])})
          case "frontier_enter":
            agg["frontier"]["N_cross"] = evt["N_cross"]
            agg["frontier"]["events"].append(evt)
          case "iss_budget":
            agg["iss"].append({"t":evt["t"], "H_bound":evt["H_bound"],
                               "frontier_linear":evt["frontier_term_linear"],
                               "frontier_switch":evt["frontier_term"]})
          case "push_Lip_check":
            pass                                             # optional: store for Section~8 audit
          case "grad_ci":
            pass                                             # Section~7 already logs CIs
          case "adaptive_radius":
            agg["drift"].append({"window":evt["window"], "Lhat":evt["Lhat"], "rho":evt["rho"]})
          case "dro_dual":
            agg["dro_dual"].append({"window":evt["window"], "eta_star":evt["eta_star"]})

# === Table builders ======================================================
function table_A_config():
    # extract unique config values from logs (band, N, hac/boot, order K)
    return {...}  # dict ready for LaTeX tabular

function table_B_domain_metrics(domain_metrics_series):
    # mean \\\\pm CI via Section~9 bootstrap helpers; effect size vs baseline
    return {...}

function table_C_stats():
    # FIM norms, HAC ellipsoid info, FDR summary
    # Build grouped p-values by (p,q) and stratum; run BH/BY
```

```
    groups = agg["fdr"]["pvals_by_group"]
    reject = hierarchical_fdr(groups)
    return {"fim_norms": agg["kappa_fim_norm"], "fdr": reject}

function table_D_control():
    pr_eps = [e["eps"] for e in agg["pr"] if "eps" in e]
    kplane_frac = mean([s["k_mode"]>2 for s in agg["svd"]])
    gap_series = [s["gap"] for s in agg["svd"]]
    N_cross = agg["frontier"]["N_cross"]
    frontier_lin = max([r["frontier_linear"] for r in agg["iss"]], default=0.0)
    return {"pr_eps_min": min(pr_eps, default=0.0),
            "pr_eps_max": max(pr_eps, default=0.0),
            "kplane_frac": kplane_frac,
            "gap_min": min(gap_series, default=NaN),
            "gap_max": max(gap_series, default=NaN),
            "N_cross": N_cross,
            "frontier_linear": frontier_lin}

# === Negative-control and degeneracy checks ============================
function check_negative_control():
    origin_hits = fraction_windows_origin_included(agg["kappa"], alpha)
    eta_med = median([d["eta_star"] for d in agg["dro_dual"]])
    return {"origin_rate": origin_hits, "eta_med": eta_med,
            "pass": (origin_hits>=0.95 and abs(eta_med)<=0.1)}

function check_degeneracy():
    k_engage = fraction([s["k_mode"]>2 for s in agg["svd"] where gap_CI_touches_zero(window)])
    chatter = fraction([a["max_angle"]>angle_thresh and not in_frontier(window) for a in agg["angles"]])
    wedin_ok = fraction_windows_principal_angles_within_band(agg["angles"], agg["svd"]) >= 0.95
    return {"k_plane_engagement": k_engage, "chatter": chatter,
            "wedin_cert": wedin_ok,
            "pass": (k_engage>0 and chatter<=0.01 and wedin_ok)}

# === Render to LaTeX tables/figures ====================================
function render_report(domain_metrics_series):
    A = table_A_config()
    B = table_B_domain_metrics(domain_metrics_series)
    C = table_C_stats()
    D = table_D_control()
    neg = check_negative_control()
    deg = check_degeneracy()
    # Write \begin{tabular}... snippets or pipe to a template engine
    return {"A":A,"B":B,"C":C,"D":D,"neg":neg,"deg":deg}
```

Guards and good practice. - *Completeness.* Fail the CI render if any table is missing its HAC/BB hyperparameters or FDR q. - *Stability.* For very small gaps (machine epsilon), force k-plane mode for a hold interval; annotate in Table D. - *Transparency.* Always print both frontier terms in the ISS row: $C_0 N_{\text{cross}}$ and $(C_0/\tau_{\min})t$.

Uses: *(Comp/Stats/Ctrl/Clos)* Section 6 PR/ISS logs; Section 7 plane selection + degeneracy logs; Section 8 FIM contraction tests; Section 9 HAC/BB/FDR; Section 4 adaptive ρ_t and $\eta^\star$.

11 Conclusion

We have developed a framework for modeling decision-objective dynamics using bigraded multivectors, cumulant expansions, and higher-level algebraic structures. Several key insights emerge:

11.1 Synthesis of Theory

- The hybrid exterior algebra formalizes independent and mixed dynamics between decision states and structural environments (G).
- Mixed cumulants $\kappa_{p,g}$ serve as rigorous markers of irreducible coupling, grounded in stochastic semantics.
- Closure can be addressed in two complementary ways: entropy-regularized variational truncation and Hopf-algebraic coproduct decompositions.
- Category theory elevates the framework, ensuring compositional semantics and cumulant-preserving functors.

11.2 Path to Computation

- A feasible computational roadmap exists: order-3 cumulant estimation with shrinkage and sketching; symbolic derivations with Python libraries; high-performance kernels in C++.
- Statistical challenges are mitigated by bootstrap confidence intervals, truncation, and Hopf-projected residuals.

11.3 Applications and Benchmarks

- Robotics: projected control improves safety/efficiency under changing constraints.
- Neuroscience: $\|K_{\text{mix}}\|$ predicts generalization in context-remapping tasks.
- Finance: $\kappa_{1,2}$ offers early warning of regime shifts.

11.4 Validation Roadmap

1. Local validity: prove and test the entropy-cumulant inequality; demonstrate stable projected control in one domain.
2. Cross-domain transfer: validate cumulant-based diagnostics in at least two domains.
3. Compositionality: demonstrate categorical functors with empirical and theoretical guarantees across interacting systems.

11.5 Vision and Future Work

- The long-term goal is to unify geometry, information theory, algebraic combinatorics, and control into a coherent methodology.
- By coupling rigorous semantics, computational feasibility, and benchmark-driven validation, the joint strategic geometry framework aspires to become a principled tool for designing and governing complex adaptive systems.

A Alignment & Normative Contracts for Joint Strategic Geometry

[Comp/Clos/Ctrl/Inf] This section specifies a *normative layer* for deploying the framework. The goal is to make *value-sensitive behavior* explicit, monitorable, and auditable using the same mathematical objects used for guarantees (Fisher metrics, cumulants, projections, PR/ISS, DRO). Philosophically: (i) treat coupling as a conserved risk resource; (ii) treat energetic passivity as a dignity floor; (iii) treat drift as epistemic debt that must be amortized; (iv) treat (co)homological obstruction as actionable *tension* rather than mere metaphor.

Operationalized philosophical primitives. Let $\widehat{\mathbf{k}}_K^{(t)}$ be the K-order mixed cumulants and FIM the Fisher information metric. Define: (i) **coupling mass** $M_t := \|\widehat{\mathbf{k}}_K^{(t)}\|_{\mathrm{FIM}^{-1}}$; (ii) **tension 2-form** ω_t as the exterior (wedge) accumulation of non-commuting evaluations (Section 8), with norm $T_t := \|\omega_t\|_{\mathrm{FIM}^{-1}}$; (iii) **colimit defect** $\Delta_t^{\mathrm{colim}}$ as a normed obstruction (Čech 1-cocycle magnitude) to forming a global state from local data.

Axioms (Alignment Layer)

1. **Safety budget (statistical coupling).** Fix a budget $B_{\mathrm{saf}} > 0$ on the Fisher-norm of mixed cumulants: $\|\mathbf{k}_K\|_{\mathrm{FIM}^{-1}} \le B_{\mathrm{saf}}$. Exceeding the budget triggers protective actions (below).

2. **Non-amplification by design.** Only functors and projections certified non-expansive in the FIM norm (Section 8) are admissible in the deployment graph; this prevents spurious growth of coupling through composition.

3. **Energetic safety.** Closed-loop designs must maintain a positive-real margin $\epsilon > 0$ on the operational band and satisfy the ISS inequality in Theorem 3 with explicit frontier term scaling $C_0 t/\tau_{\min}$.

4. **Caution under drift.** Radius policy (Definition 5) with $\rho_t = c\,\widehat{L}_t\,\tau$ is used; δ is set from finite-sample W_1 rates (Remark 2). Dual $\eta_t^\star$ is logged and bounded.

5. **Transparency and consent.** Symmetry choices (whitening, invariance groups), truncation order K, PR bands, and statistical CI/FDR methods are disclosed wherever decisions impact agents.

6. **Subsidiarity & reversibility.** Prefer interventions that (i) act at the smallest locus compatible with safety, and (ii) admit an explicit rollback map $\mathcal{R}$ with $\|I - \mathcal{R}\circ\mathcal{A}\|_{\mathrm{FIM}} \le \beta$ for a set $\mathcal{A}$ of admissible actions and small $\beta > 0$.

7. **Exposure accounting & hysteresis.** Maintain a cumulative exposure integral $E_t := \int_0^t M_s\,\mathrm{d}s$ with budget B_{exp}; apply hysteresis bands on triggers to avoid policy thrashing.

8. **Tension budget (coherence).** Fix a budget B_{ten} on $T_t = \|\omega_t\|_{\mathrm{FIM}^{-1}}$ and a defect budget B_{col} on $\|\Delta_t^{\mathrm{colim}}\|$. These capture strategic vorticity and consensus obstruction, respectively.

9. **Least privilege & enumerated authority.** Policy knobs (below) are namespaced and scoped; each actuation is justified by a minimal-sufficient trigger and recorded with a capability token.

10. **Value elasticity bounds.** For declared objectives J, estimate Fisher–Riesz sensitivities $\|\nabla J\|_{\mathrm{FIM}^{-1}}$ and enforce a Lipschitz envelope so that interventions do not induce *surprise* outside declared bounds.

Normative control policies (trigger–response)

Let $M_t := \|\widehat{\mathbf{k}}_K^{(t)}\|_{\mathrm{FIM}^{-1}}$, ϵ_t the PR margin, $\eta_t^\star$ the DRO dual, $T_t := \|\omega_t\|_{\mathrm{FIM}^{-1}}$, $\Delta_t^{\mathrm{colim}}$ the colimit defect, and E_t the exposure integral.

Trigger T1 (statistical over-coupling).
If $M_t > B_{\mathrm{saf}}$ (with CI excluding B_{saf}), then: (i) apply Dirac Π (if not active); (ii) reduce controller gains via a monotone map $g(\cdot)$ constrained to the admissible PR set; (iii) increase truncation (reduce residual) by lowering K's residual penalty threshold.

Trigger T2 (energetic risk).
If $\epsilon_t < \epsilon_{\min}$ on the band, revert gains to a safe envelope and widen Oustaloup order/band as per Lemma **??**/10.

Trigger T3 (distributional drift).
If $\eta_t^\star > \eta_{\max}$ for a persistence window, enlarge ρ_t (cautious inference) and/or shorten τ; optionally halt adaptation within frontier tubes.

Trigger T4 (tension flare).
If $T_t > B_{\text{ten}}$ or $\|\Delta_t^{\text{colim}}\| > B_{\text{col}}$, then: (i) decompose ω_t along certified non-expansive projectors to locate sources; (ii) prefer *enzyme* actions $\mathcal{E}$ (Definition below) that reduce T_t with bounded control effort; (iii) freeze cross-component functors that break commutativity until tension recedes below the lower hysteresis band.

Trigger T5 (excess exposure).
If $E_t > B_{\text{exp}}$, enforce a cool-down: suspend non-essential adaptations, raise $\epsilon_{\min}$ by a margin $\kappa > 0$, and restrict authority to a minimal capability set.

Trigger T6 (consent/coherence failure).
If transparency feedback indicates dissent or if a coequalizer certificate fails (no consensus colimit), then: (i) fall back to local policies (subsidiarity); (ii) require an override (see Governance) with a justificatory taxonomy entry and an ex-post harm audit.

Definition (Catalytic policy actuators, "enzymes"). An actuator $\mathcal{E}$ is catalytic if it monotonically reduces T_t while preserving (i) the PR margin and (ii) coupling mass budget: for some $\alpha, \gamma > 0$, $T_{t+1} \leq T_t - \alpha\|u_t\|$ and $M_{t+1} \leq M_t + \gamma\|u_t\|$ with (α, γ) certified and $\epsilon_{t+1} \geq \epsilon_{\min}$. Catalog and reuse $\mathcal{E}$ with provenance.

Accountability artifacts

Per deployment episode, publish: (i) windowed M_t with HAC or bootstrap CIs and FDR settings (Section 9); (ii) PR margin ϵ_t range; (iii) frontier counts and the ISS frontier term $C_0 N_{\text{cross}}(0,t)$ and $C_0 t/\tau_{\min}$; (iv) $\widehat{L}_t$, ρ_t, $\eta_t^\star$ trajectories (Section 4); (v) symmetry/whitening invariance checks (Assumption 3); (vi) tension metrics T_t and $\|\Delta_t^{\text{colim}}\|$ with localization plots; (vii) exposure E_t and hysteresis bands; (viii) an *override ledger* (see below) with taxonomy codes and outcomes; (ix) a reproducibility capsule (non-expansiveness certificates, seeds, and audit scripts) to replay ISS frontier accounting.

Governance hooks (interfaces)

The alignment layer exposes:

- **Policy knobs:** (B_{saf}, $\epsilon_{\min}$, q_{FDR}, K, c, τ, B_{ten}, B_{col}, B_{exp}) with scope (global vs windowed) and provenance. Each knob carries a capability token and an expected effect size range.
- **Override contracts:** An operator may override T1–T6 but must attach a justification chosen from a fixed taxonomy (safety-of-safety, humanitarian, legal, fiduciary, experiment), cite the triggering metrics, and specify an ex-ante rollback map $\mathcal{R}$. Overrides are logged and included in accountability artifacts.
- **Audit API:** An external auditor can recompute M_t, verify non-expansiveness certificates (Section 8), replay ISS frontier accounting, and independently compute T_t and Δ_t^{colim} from provided local evaluations.

Remark (Ethical posture). The alignment layer encodes a conservative stance: prefer reduced coupling (smaller $\|\mathbf{k}_K\|_{\text{FIM}^{-1}}$), preserved energetic passivity (PR margin), cautious inference under drift (larger ρ_t), and minimized tension (smaller T_t and $\|\Delta_t^{\text{colim}}\|$) when uncertainties grow. Choices are explicit and tunable, with rollback maps and hysteresis to guard against over-correction. The same mathematical guarantees support domain-specific ethics without appeal to untestable narratives.

Uses: Thm. 3, Lem. **??**, Lem. 10, Def. 5, Rem. 2, Lem. 12.